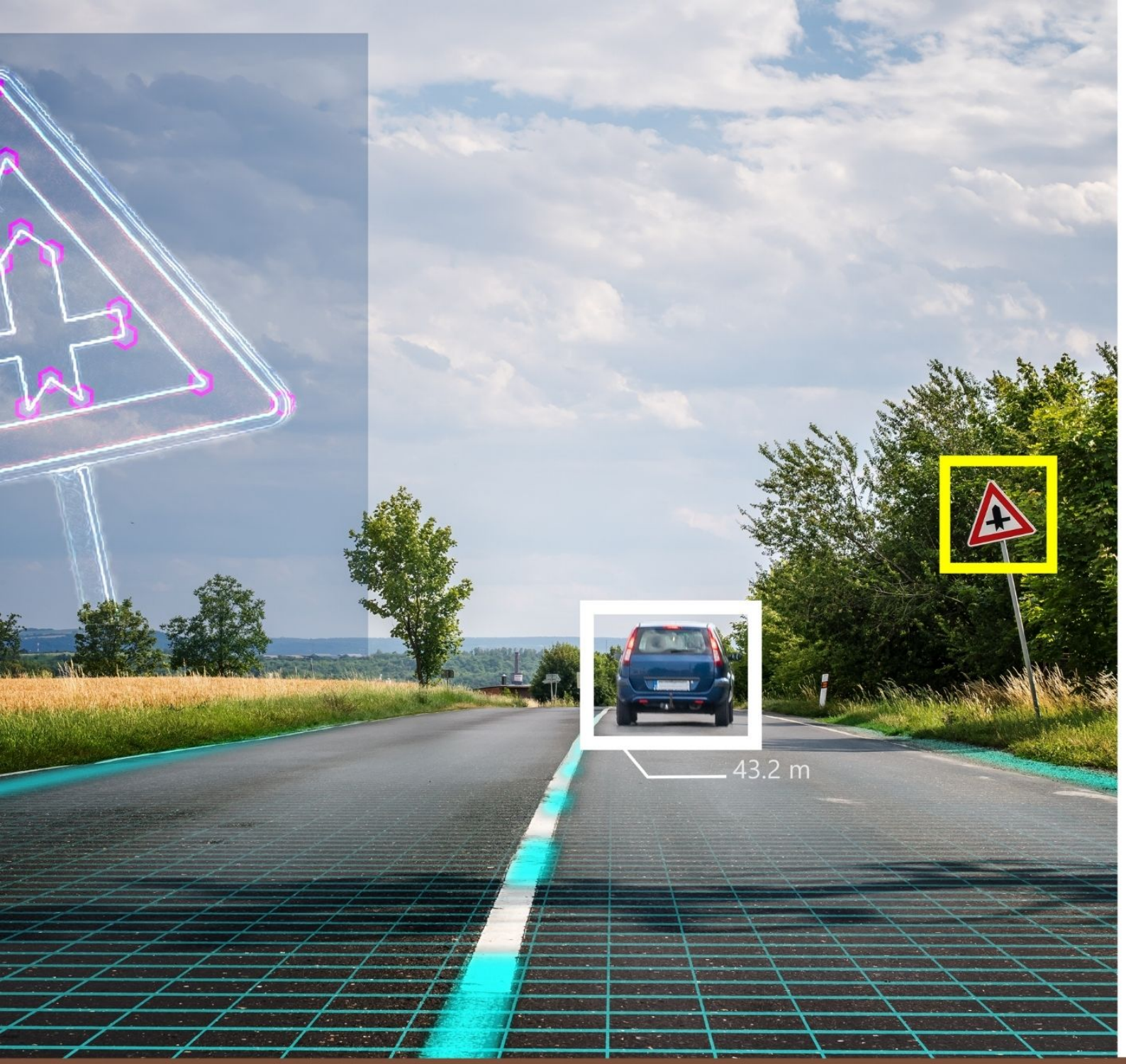


COMPUTER VISION

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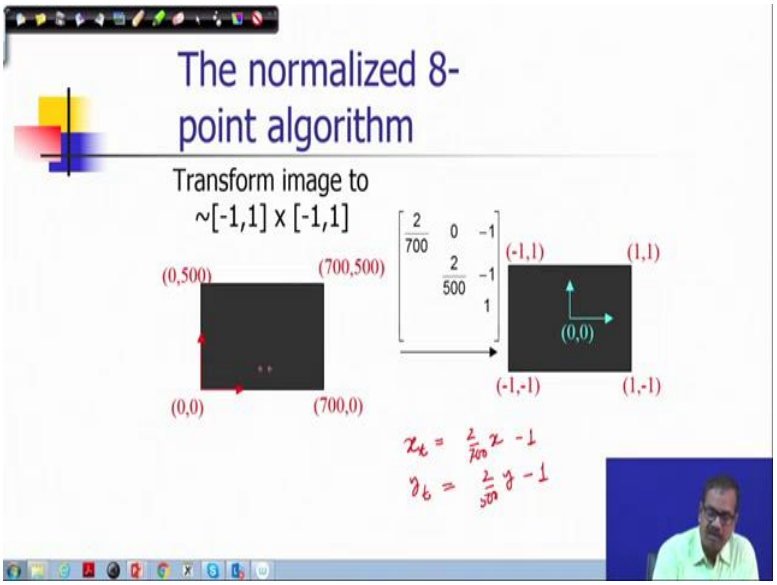
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Computer Vision
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Lecture - 20
Stereo Geometry Part – V

We are continuing our discussion on estimation of fundamental matrix and in the last lecture we discussed how the equations can be formed using the pairs of corresponding points in relation to the elements of fundamental matrix. And, by solving those equations you can get those elements of fundamental matrix. One of the things we are discussing that we require to perform coordinate transformation as the dynamic ranges of coefficients of those equations they widely vary because of the nature of the equations.

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So, we will continue that discussion and as we discussed that we will map the coordinate points to these ranges and bring them into these ranges. An example is shown here. Usually that is a very common case that no image coordinates vary between say 0 to 700 in this example and 0 to 500 there are other values. So, it is a typical range and there you can see product of those coordinates will have a very large dynamical range compared to single variables of coordinate points, single points. So, the example of a transformation will convert these coordinates into these ranges as given here.

$$Transform\ image\ to \sim [-1,1]X[-1,1]$$

You can see that this transformation is given in this form, that if I multiply this matrix with x', I will get the transformed point.

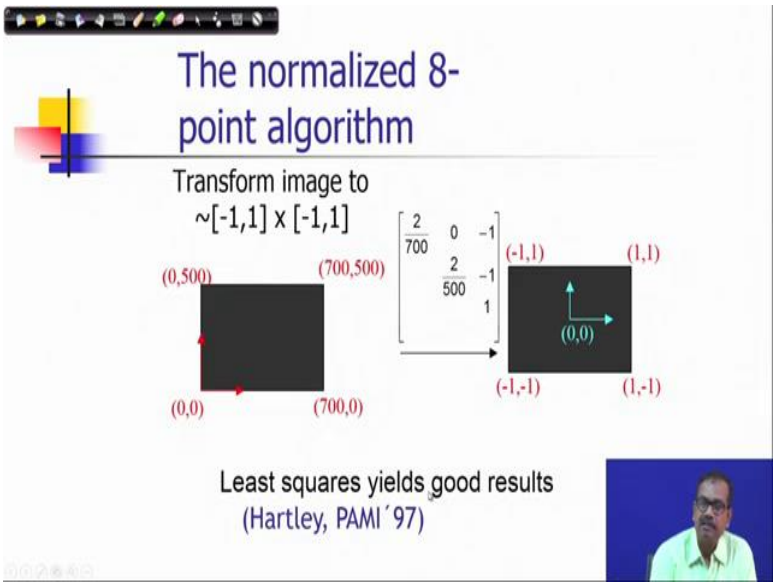
So, let me denote the transform point as say xt coordinate; that means,

$$x_t = \frac{2}{700}x - 1$$

$$y_t = \frac{2}{500}y - 1$$

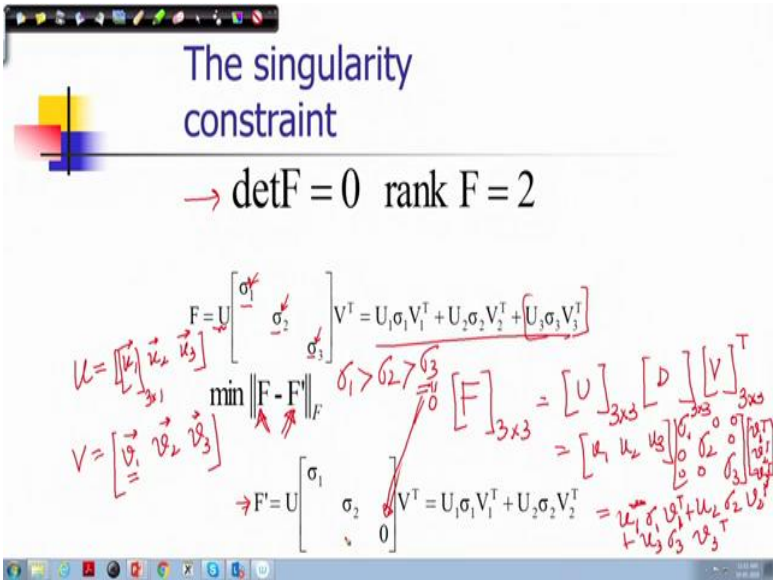
So, given (x, y) will be converted into this value. So, that would translate the origin to the (0, 0) in the middle of this (-1, -1) has disc affect. Anyway as discussed that you have to apply the corresponding computations to apply that transformation back into what you get as a fundamental matrix out of this computation. Then the inverse transformations are to be applied to get their original fundamental matrix in the respective coordinates of the image planes.

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We will now continue, we will see what other aspects are there. So, as we mentioned that this is a paper where actually this particular technique is discussed and it has been shown that least square error method yields good results when we perform this transformation.

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So, now we will consider the other aspect of this estimation, I mentioned that in the previous case that we have not applied the constraint of singularity on the fundamental matrix. So, the solution what you get that is not truly giving you a fundamental matrix unless it satisfies that singularity constraints. So, how to enforce that singularity constraint? So, that is what we will be discussing here. So, here you see that this constraint is expressed in the form of that the determinant of the matrix F should be equal to 0. And, also we know since its rank deficient, it should be less than 3 and in this case it would be 2.

$$\det F = 0 \; and \; rank(F) = 2$$

So, there is 1 zero particularly of this fundamental matrix we discussed also, that there is a epipole that is left a epipole. Epipole in the reference image reference plane image plane that is a 0 of this fundamental matrix that is a right 0. Similarly, there is a left 0 which is a right epipole; Now if I perform singular value decomposition of a fundamental matrix. So, we present singular value decomposition in this form. So, you see that U is a matrix which is orthogonal or we can make it also orthonormal which means if I take the column vectors.

$$U = [u_1 \; u_2 \; u_3]$$

So, let me represent U as a set of column vectors in this form, I will use this notation. So, each vector is a column vector which is of 3 cross 1 dimension; that means, U is a 3x3 matrix.

And, in singular value decomposition all these column vectors would be orthogonal or you can make it orthonormal also as I mentioned. Similarly, V is also another matrix whose column vectors are also orthogonal. So, you have V1, V2, V3 so, this is also orthogonal. So, any square matrix or even any matrix not only square can be decomposed into this form, into this particular structure in using singular value decomposition. So, in this case particularly I will discuss with respect to square matrix only for in this context so, it can be decomposed into this form.

$$F = u_1\sigma_1{V_1}^T + u_2\sigma_2{V_2}^T + u_3\sigma_3{V_3}^T$$

So, U as I mentioned 3x3 then there is a diagonal matrix, and then V which is also a matrix, but we have to take transpose of this V. So, I can write also like [u1 u2 u3] and this diagonal matrix there in this case it is also a 3x3 matrix. You can see that if I multiply I will get a 3 x3 matrix. So, diagonal matrix as we know that all off-diagonal elements will be 0 except the diagonal elements which should be non-zero not necessarily, that it should be non-zero. It could be 0 also and we can apply no sign laws it is, it we can adjust the signs in both the sides so, that no one can make it all positive.

So, if I shuffle these columns similarly shuffle the rows of [v1 v2 v3] in the same order still we will get the same matrix F. So, I will shuffle in such a way that

$$\sigma_1 > \sigma_2 > \sigma_3$$

which means I will take those column vector those pairing U1 and V1 whose sigma is those singular value. So, σ_1 this is called as singular values whose singular value is the maximum here, then the next maximum then the minimum. So, this is how the singular value decomposition works and this is how you can see the matrix can be considered as a super positions of 3 rank 1 matrices.

So, you are just summing up 3 rank 1 matrices so, each one is a 3x3 matrix. So now, what you can do that you have to estimate a singular 3x3 matrix which should be very close to your estimated F matrix, now that is a computational problem. So, F you have estimated

from the data, but that is not singular. So, your objective is to get that solution F' which should be very close to your estimated F which means you have to minimise the Frobenius norm.

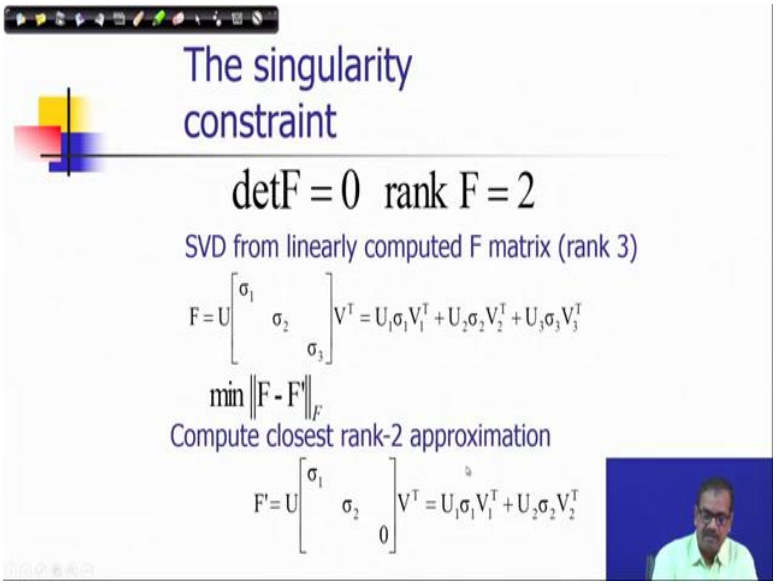
That means, element wise if you take the square of their differences and add them and take the square root of the sum that is a Frobenius norm. It is as good as saying that this is L2 norm when you are representing a matrix in the form of a vector by concatenating all its rows or its columns in whatever way. So, the solution lies in the fact that this should be minimum that theory says that when you set the minimum singular value to 0; that means, you are setting to 0 so, you are ignoring this part. So, once you set it 0, now then this F prime is a singular matrix because that is one of the conditions one of the signature of a singular matrix.

That if you perform singular value decomposition, then at least one of the singular value should be 0 and if I reorder them all those singular values in order of their increasing values. And, as I mentioned that we enforce that singular values should be positive we can enforce positive point 0 so, the minimum values should be 0 in that set. So, in this case number of 0s or number of non-zero singular value will also provide us the rank of that matrix. So, since the rank of this fundamental matrix is 2 so, we are keeping two non-zero singular values and only setting the minimum singular value to 0.

So, this becomes your solutions now and you have applied the singular constraint in this fashion. So, this is that we are computing singular value decomposition.

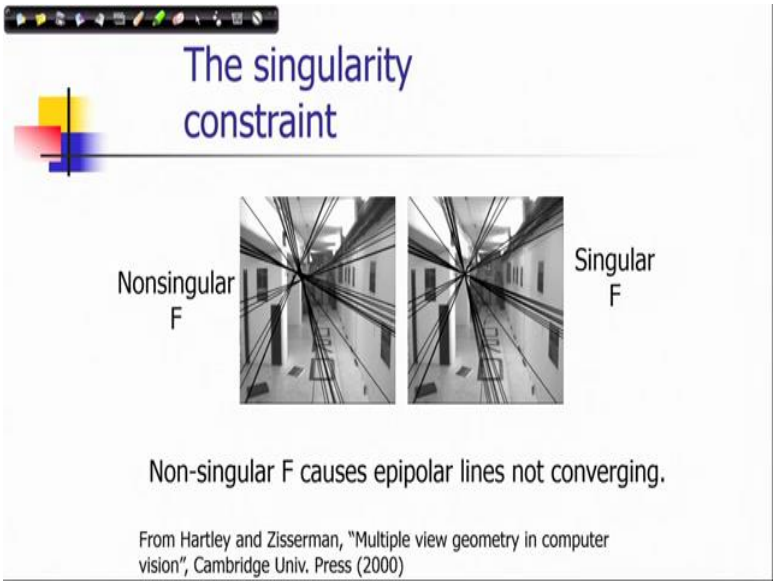
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$$Closest\ rank\ 2\ approximation, F' = u_1\sigma_1{V_1}^T + u_2\sigma_2{V_2}^T + u_3\sigma_3{V_3}^T$$



And this is the closest rank 2 approximation from that estimated F matrix.

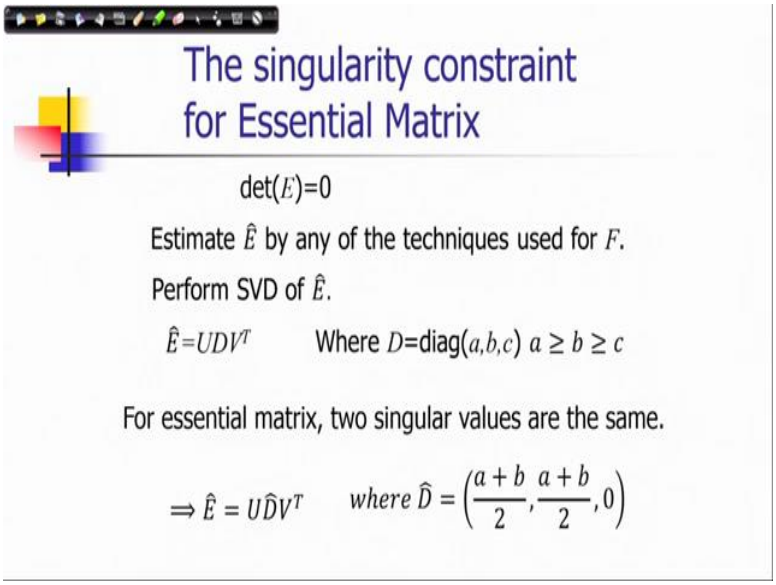
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So, one of the nice example that I could get from the book of you know Hartley and Zisserman that is "Multiple view geometry in computer vision". You can see there are the effect of singular F and non-singular F by showing how epipolar lines behave in a particular image plane. So, this is a right image plane with respect to a left image plane which is a reference plane and we are drawing the epipolar lines of all the points corresponding points of the left image planes in the right image plane.

It is expected that all epipolar lines should be converging to a single point and that is the right epipole of that stereo system. So, you can see that if it is non-singular then they are not converging. Only their convergence is ensured if it is a singular. So, that is why geometrically also this property needs to be satisfied and that is why this constraint is so important in this respect.

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Now, if I consider the estimation of essential matrix there is another interesting feature, I told earlier also the number of independent parameters in essential matrix is 5. So, which means there are few more constraints here. So, let us find out what are those constraints.

$$\det(E) = 0$$

So, just remember that essential matrix is the same fundamental matrix when your camera is calibrated which means the calibration matrix of the camera. And, you can convert all the image coordinates into the canonical form and express the fundamental matrix just by using the rotation and translation of the word coordinate to camera coordinate system those parameters. So, extrinsic parameters are the only constraints and since there are 6 parameters and also the scale is another constraint so, there will be 5 independent parameters in this case. So, suppose you have a calibrated camera; that means, you know the calibration matrix. So, you can take and make that conversions and then you apply the same estimation technique. And, whatever fundamental matrix you have estimated that is essentially an essential matrix because, you get this canonical coordinates and you get in

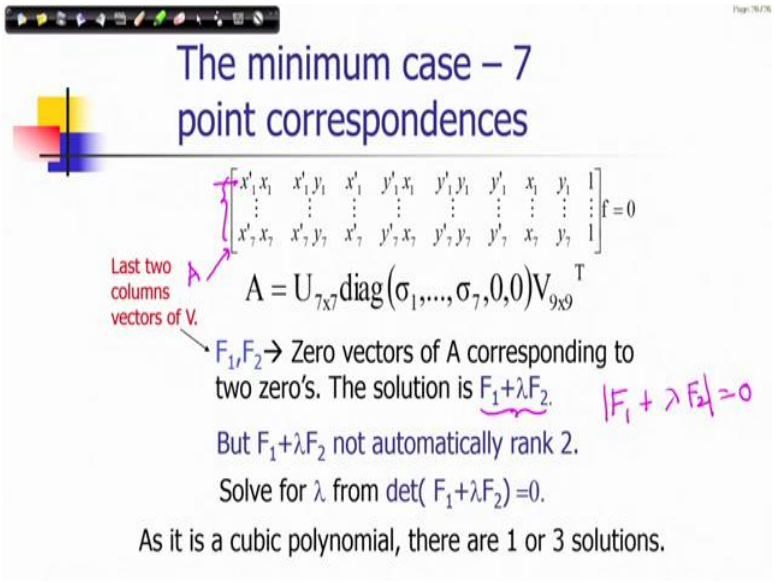
that form of the fundamental matrix. So, there also we need to perform singular value decomposition to enforce the constraint of singularity.

But, there is one interesting property of essential matrix that the two singular values that you get here should be also equal.

$$\hat{E} = U\hat{D}V^T \, , \hat{D} = (\frac{a+b}{2}, \frac{a+b}{2}, 0)$$

So, the way you can enforce is that you consider the average of those two values (a, and b) and set them to the σ_1 and σ_2 in my previous technique whatever we discussed. You just make these modifications when you are estimating an essential matrix.

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Let us discuss now how to estimate fundamental matrix using 7 point correspondences which is the minimum number that is required as we mentioned earlier. So, if you have 7 point correspondences there would be 7 such equations involving the elements of the fundamental matrix as we also discussed in our previous slide. So, we can see that each row is also formed in the same way what we considered earlier. And, there are 7 such row corresponding to each of the 7 pairs of corresponding points. So now, if you would like to solve this as you can see this is a set of homogenous equations and since the number of elements in F is 9 and you have 7 equations.

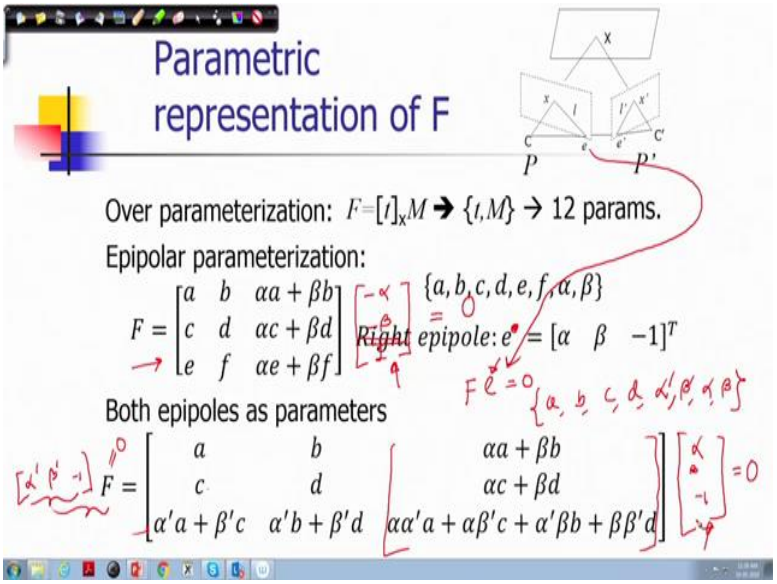
So, the rank of this matrix is 7 and here we can use the singular value decomposition of this matrix A. So, we consider this matrix is A and if I perform a singular value decomposition then we will be naturally expecting there will be two 0s in this case. And, the U and V you can consider know their dimensions in this form and V there will be corresponding 0 vectors of this particular matrix A. So, those 0 vectors are the last two columns of V and they can give you the solutions of the equations. Because, 0s are the solutions of A are they are the solutions of these equations and they are the corresponding fundamental matrix.

But since there are two such zero vectors so, actually the solution is a linear combination of these two zero vectors. So, we can express the solution as a linear combination of two zero vectors F 1 and F 2 as it is shown here and this linear combination is given in this form. So, where lambda is any scalar constant, but you have to determine exactly which linear combination which lambda value.

$$|F_1 + \lambda F_2| = 0$$

Because, the fact is that the determinant of this should be equal to 0, it should be rank deficient matrix which means if I take the determinant of this matrix, if I take the determinant that should be equal to 0. And, since the determinant it is not automatically rank 2 we have to solve for lambda which will give you the 0. And, it will be a cubic polynomial because it is a 3x3 matrix and in cubic polynomial we would like to get a real valued solutions of lambda. And, there are two situations either there will be one real valued solutions or there will be three such solutions. So, a maximum we have three such possibilities of fundamental matrices in this case that is a technique we discussed.

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So, the essence is that you have to perform singular value decompositions or rather eigen decomposition of the A^T matrix here. And, and solve for the lambda which is the coefficient of linear combination of the eigen vectors by enforcing the determinant of the combination linear combination should be equal to 0. So, let us consider now the other aspect of the fundamental matrix. So, how fundamental matrix can be represented in different parametric form. So, we have already seen how many number of independent parameters should be there, but not necessarily your representation will have that intrinsic characteristics.

So, it you can represent with more number of parameters, but then you have to enforce the constraints; that means, the independencies among those parameters needed to be enforced there. So, in the estimation process and this is not always ensured and a post estimation those operations are taken care of as we discussed that how we can make the determinant 0 for F. So, over parameterization representation could be in this form; that means,

$$F = [t]_X M \rightarrow \{t, M\} \rightarrow 12\ parameters$$

$$e = [\alpha\ \beta\ -1]^T$$

where t is a epipole. And, in this form you see that there are more number of parameters like you have epipoles t and M.

$$F = \begin{matrix} a & b & \alpha a+\beta b \\ c & d & \alpha c+\beta d \\ e & f & \alpha e+\beta f \end{matrix}$$

So, M is 9 and t is a 3x1 so, we can write 12 parameters. Now, we know essentially there are seven independent parameters. One of the interesting representations which follows this parameterization is this one, you can see that how many independent parameters are there a c e b d f 6 then alpha and beta. So, there are 8 independent parameters. The interesting part is that the singularity constraint on f; that means, the determinant that is equal to 0 here because if I take the linear combination of these two columns; the third column is a linear combination of these two columns.

$$F = \begin{matrix} a & b & \alpha a + \beta b \\ c & d & \alpha c + \beta d \\ \alpha' a + \beta' c & \alpha' b + \beta' d & \alpha\alpha' a + \alpha\beta' c + \beta\alpha' b + \beta\beta' d \end{matrix}$$

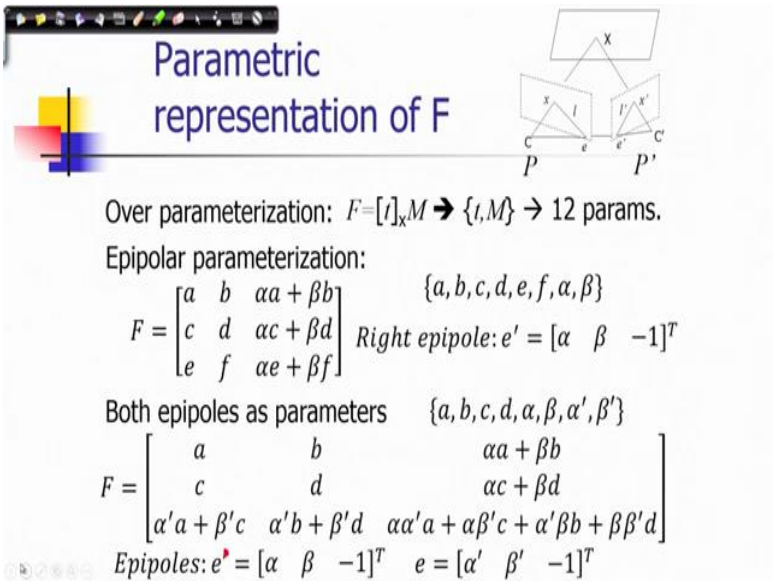
So, we know that in this case this is a rank deficient matrix and one of the column is dependent of others. So, also its determinant should be equal to 0 that is the property of determinant. And, if I multiply this F say with$[-\alpha - \beta\ 1]$, you will find that you are getting 0. So, you perform minus alpha a minus beta b plus alpha a beta b that is 0 and in this way. So, what does it mean? It means that the left epipole; that means, 0 of F is this 1 so, which means it is an epipole. So, alpha and beta has an interpretation in this form or you can see alpha beta minus 1 so, it is because you are multiplying F e should be equal to 0.

So, I think this is fine. So, it should not be right epipole e, it should be the left epipole e it should be this one ok. So, let us consider the next one that is both epipoles as parameters. So, you have this number of parameters a b c d e f alpha beta. Now, this is another representation you can see that how many number of parameters you are having here. You have a b c d then alpha prime beta prime and alpha and beta; that means, here also you have 8 parameters a b c d alpha prime beta prime and alpha beta. So, a b c d alpha prime beta prime alpha beta.

Now, in this case you can find out that if I multiply F with once again with alpha beta minus 1. So, here also you can see that the third column; that means, this column this is a linear combination of first two columns which is alpha into this and beta into this will give me the third column. And, alpha in yeah, and also the third row is a linear combination of the first two row which is alpha prime into first row and beta prime into second row will me the third row. So, if so alpha beta minus 1 that should be equal to 0 like the previous one. And, then if I perform on the other way if I perform the multiplication alpha prime

beta prime minus 1 with F that should be also equal to 0; that means, this becomes your right epipole and this becomes your left epipole so, in this set up.

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So, once again now we have the same problem of representation. So, e should be alpha beta minus 1 and e prime should be so, here also you have a problem. So, please note this corrections e should be alpha beta minus 1 and e prime should be alpha prime beta prime minus 1 as per my diagram.

So, let me stop here and we will continue our discussion in the next lecture.

Thank you for your listening.

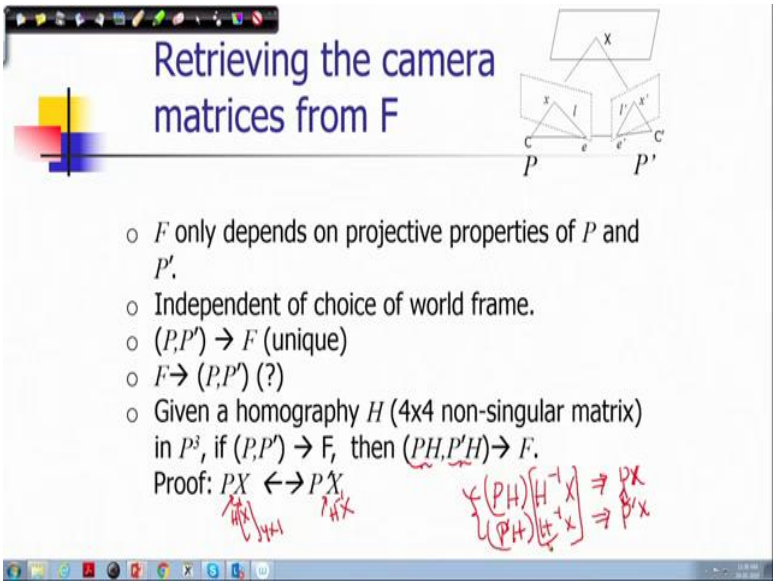
Keywords: fundamental matrix, singularity constraint, essential matrix, parametric representation.

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Lecture – 21
Stereo Geometry Part – VI

We are discussing about fundamental matrix of a Stereo Geometry and we have seen how fundamental matrix plays a very distinctive role in characterizing its geometry. And in the last lecture, we discussed how fundamental matrices could be estimated given the observations of corresponding pairs of points and also some parametric forms of representation of fundamental matrix which also sometimes helps us in formulating the estimation problem.

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Now, in this lecture, we will be discussing that how we can retrieve the camera matrices from fundamental matrix F. So, note that in our previous problems, we have not considered estimation of camera matrices. If we know the camera matrices, then computation of fundamental matrix is very direct; it's very simple. We will see here the reverse is not the same.

So, let us consider these aspects that if that is a fundamental matrix, it only depends on the projective properties of P and P'. So, given a pair of P and P', you get an unique F and the interesting part is that it is independent of choice of world frame means world coordinate

system. That means, if I have a stereo setup and I can move this set up by keeping its relative orientation the same; relative orientation of the image access the same.

I can move this stereo setup in at any point of the world frame whatever may be my world coordinate convention here. Still I should have fixed world coordinate system and then, I can move the stereo system at any point, still we will get the same fundamental matrix which is not true for projection matrix. If my world coordinates also move and based on that the corresponding points of image coordinate and world coordinate points will change. Because world coordinate have been transformed then your projection matrix will also get transformed, but this is not true for fundamental matrix. It is solely dependent on the image coordinates.

So, if you have same pairs of image coordinates for the corresponding pairs of points, we will get the same fundamental matrix. As you know that even if you move the cameras at any point in the imaging system, a coordinates will coordinates will vary, but still fundamental matrix will remain the same. So, this is one of the interesting property. So, P P prime gives a fundamental matrix F it is unique.

So, the question is that whether we can get the projection matrices P and P' from a fundamental matrix F. Now, this is a property which we can easily verify and by which we can see that we do not have a unique solution in this case. That means, though projection matrix is a pair of projection matrices P and P' give an unique F, but there could be other pairs of projection matrices which will can give me the same fundamental matrix F.

So, you consider the situation that there is a homography matrix in the form of 4x4 homography which means is a linear transformation. So, if I multiply P with a 4x4 nonsingular matrix which is invertible matrix. So, from PH, I can again get P.

$$If\ (P, P') \rightarrow F, then, (PH, P'H) \rightarrow F$$

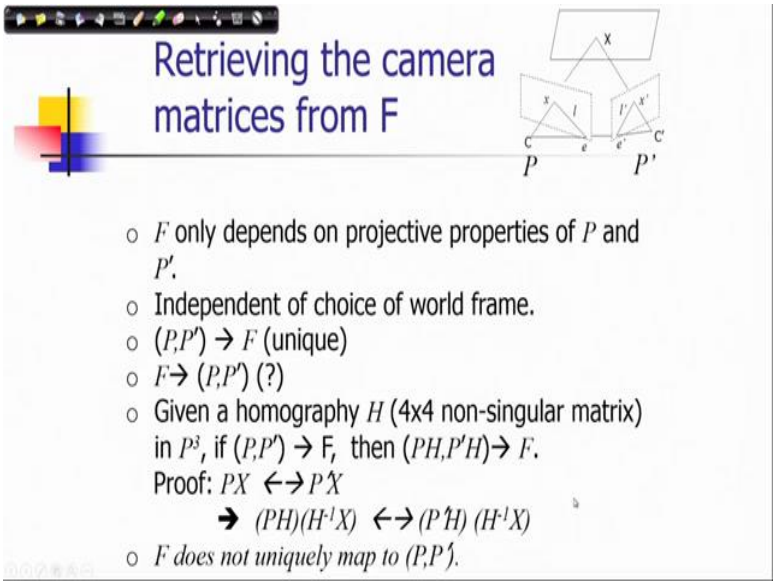
There is no problem, I can simply multiply with its inverse, but the thing is that if I multiply PH; H with this pair and P' with H, then I will get the same F. That is a theory.

So, the proof is that. So, if I consider a particular scene point X and consider this stereo system, then the corresponding pair of points will be PX and P'X. Now, consider a

transformation of scene point by this homography 4x4 matrix; that means, by a 4x4 nonsingular matrix a linear transformation. So, you know that X is a 4x1 column vector. So, if I multiply with 4x4, it still remains a 4x1 column vector. So, now, you consider rather I will make H^{-1} instead of H, let me do it H^{-1}. H^{-1} will be also a 4x4 matrix because I mentioned H is an invertible matrix.

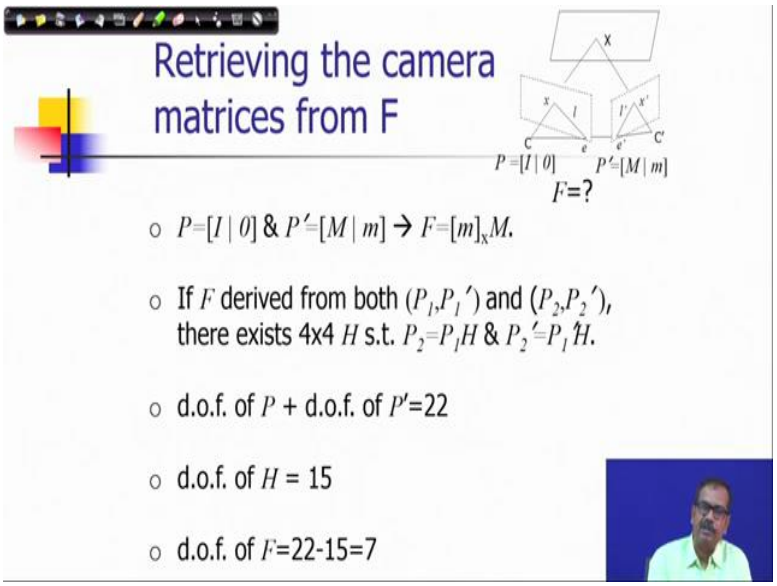
So, if I consider another projection matrix PH and multiply with $H^{-1}X$, we will get PX the same points. Similarly, P'H multiply with $H^{-1}X$, we will get P'X which are the same corresponding pairs of point. So, same corresponding pairs of points have these solutions. It could give me a fundamental matrix satisfying no a set of projection matrices in the form of PH P'H and corresponding scene points as $H^{-1}X$. So, this is what this proof is and that is I can summarize once again just to make it a clean display.

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So, this is just we derived; this is the scenario. So, the summary is that F does not uniquely map to P'.

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So, once again we continue this discussion. So, let us consider this particular scenario, we discussed earlier that projection matrix is in the form of a canonical from that is for the reference camera which is given by this

$$P = [I|0], P' = [M|m] \rightarrow F = [m]_X M$$

Then we know that given this pairs of projection matrices, its fundamental matrix of the stereo system can be computed using this expression.

Note that this m is a right epipole; that means, this is column vector is the right epipole of this. So, this M provides you also the homography at infinity. So, we note e' cross homography at infinity that gives me a fundamental matrix that is a relationship we are discussed here.

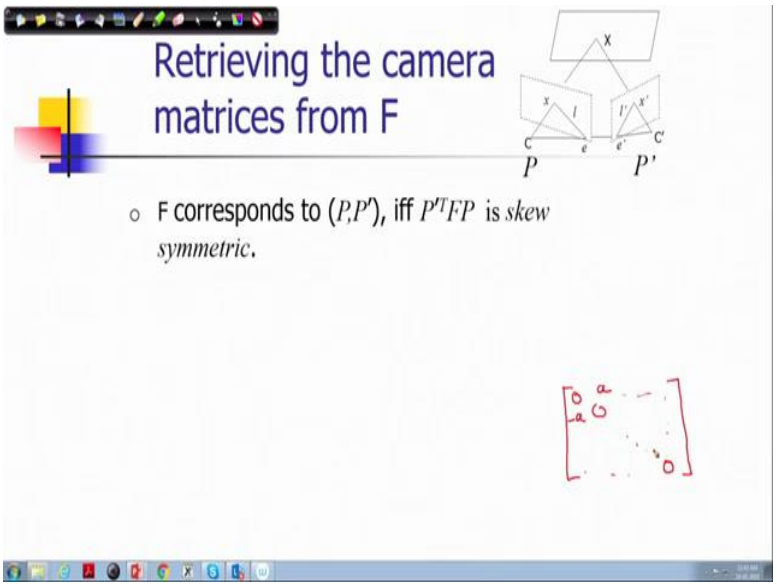
So, we discuss that if we can derive F; if you get F from two camera systems the same F, then there should be a relationships between the projection matrices of these two camera systems. It collaborates with the theory, what we discussed in the previous slide. And we can explain that because of this constraint, we can explain how degree of freedom of F is related to the degree of freedom of projection matrices.

Say we know that number of independent parameters of a projection matrix is 11. Just the scale that it should be scale equivalent that is all. So, there are 12 elements and so, number of unknowns are 11 or number of independent parameter is 11. So, since there are in a

stereo system, it involves two projection matrices. So, maximum degree of freedom could be 22, but since they are related by this homography 4x4 homography; once again homograph 4x4 homography matrix is also a projective element. That means, there is it is described upto scale.

So, one of the element could be a scale element. So, there are 15 degree of freedom of H. So, that is the constraint imposed on a fundamental matrix that given that fundamental matrix, projection matrices should be related given any reference projection matrix. So, 22-15 will give you the degree of freedom of F. So, this justify how that degree of freedom of a fundamental matrix is 7.

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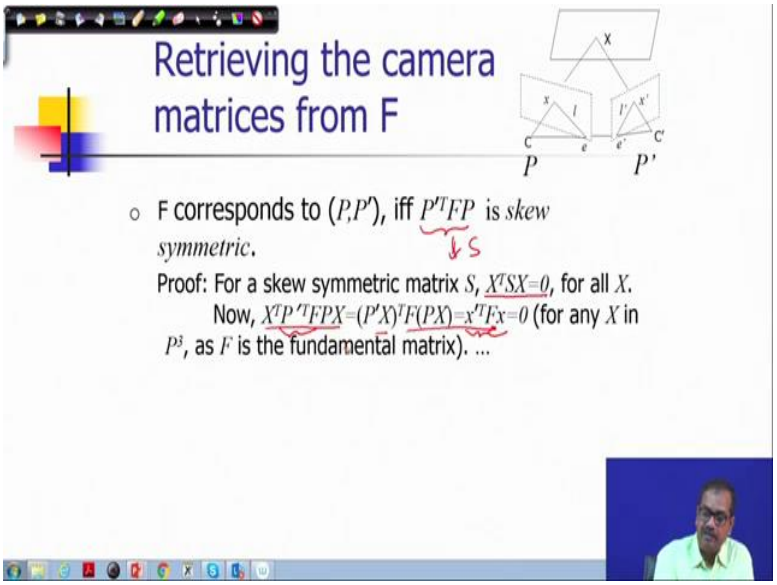
There is another interesting fact that we like to discuss that the fundamental matrix corresponds to a projection matrix P and P' if and only if $P'^{T}FP$ is a skew symmetric matrix. That means, it check the compatibility. So, if I give you a fundamental matrix and if I give you P and P', immediately I can check with this relationships whether they are compatible or not.

I need not compute F from P and P' to check it. I have to check the scale equivalence there of course, but in this case simply I have to check whether $P'^{T}FP$ is skew symmetric.

So, what is skew symmetric? Just to remind you, a skew symmetric is a matrix where the transpose of this matrix should be the negative of, this one and the diagonal element would be 0. So, this is one example.

$$A^T = -A$$

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So, just to prove this fact one of the interesting property of skew symmetric matrix that is exploit that if you consider any skew symmetric matrix, then this fact is true; that means

$$X^T SX = 0 \; for \; all \; X$$

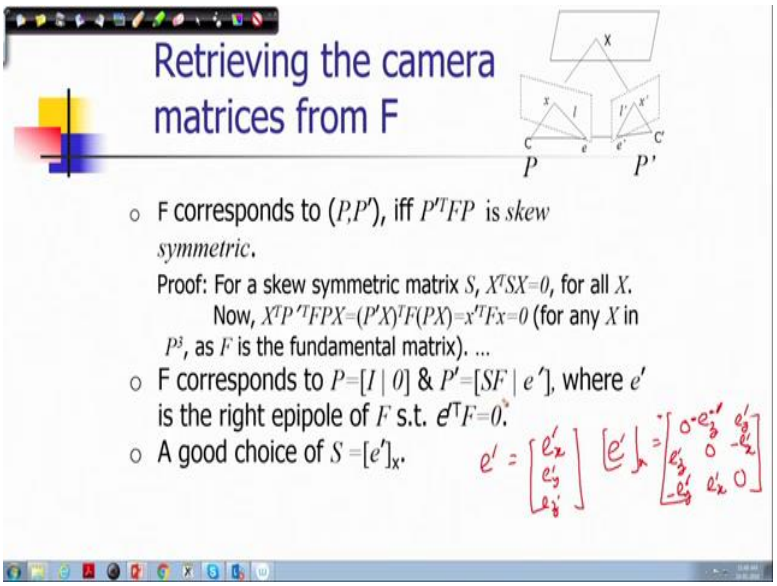
You have to choose a proper matrix dimension. So, a skew symmetric matrix has to be a square matrix and when it is true for all X. So, if it is a skew symmetric matrix, it has to be true for all X because of that negation properties that the transpose of this is a negation.

So, now, we will be showing that if I perform this operation,

$$X^T P'^T FPX = (P'X)^T F(PX) = X'^T FX = 0$$

You see that this is true for only the corresponding pair of points, but for any scene point in the three-dimensional phase or in the homogeneous coordinate four-dimensional homogeneous coordinate space, this is true and that is how if this quantity $P'^T FP$; this is a skew symmetric matrix.

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Then another relationship we will be knowing here that F corresponds to P and P', where P' is given in this form. That means, one of them is a skew symmetric matrix; S is a skew symmetric matrix and F is a fundamental matrix.

So, this relation is giving me a particular pair of no projection matrices. It is helping us to form at least a pair of matrices, where which will give me the corresponding F. So, I will not give the details of the proofs, but these results are interesting. So, that we will be writing here. So, one of the good choice of a skew symmetric matrix, we have already discussed that is e prime cross.

We know $[e']_X$ is a skew symmetric matrix and if you remember that if I consider e', if I represent e' as say

$$e' = \begin{matrix} e'_x \\ e'_y \\ e'_z \end{matrix}$$

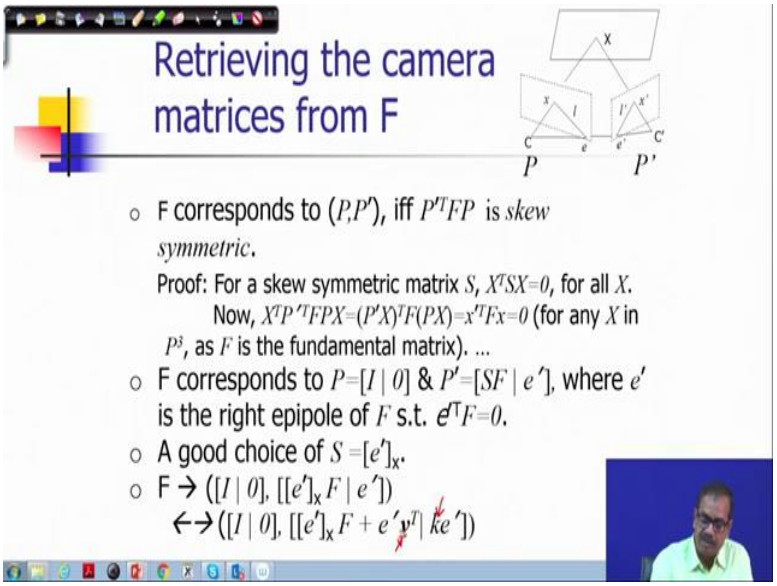
then $[e']_X$ is given by a skew symmetric matrix in this way.

$$[e']_x = \begin{matrix} 0 & -e'_z & e'_y \\ e'_z & 0 & -e'_x \\ -e'_y & e'_x & 0 \end{matrix}$$

.

So, you see that this is a skew symmetric matrix. So, the solution from F at least we can get one set of one pairs of projection matrices; one of them is in the canonical from [I|0] and the other one would be$[[e']_X F|e']$. So, this is at least a solution you can get from here.

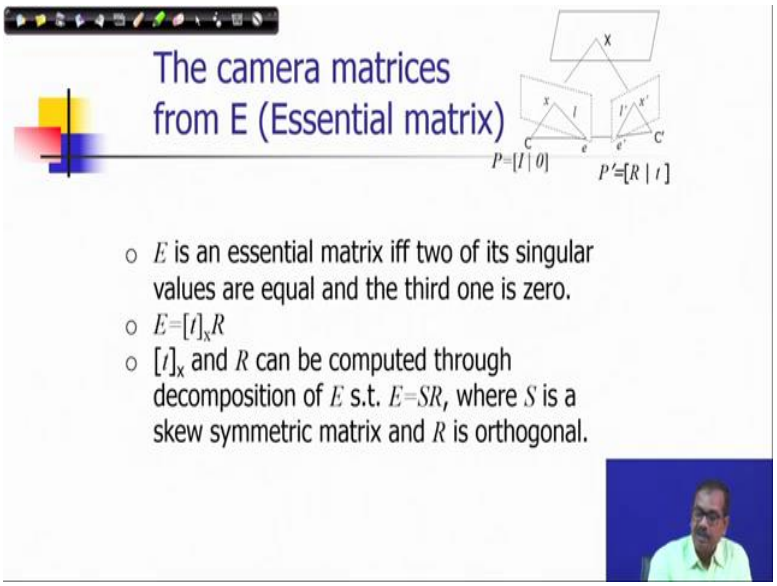
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So, this is what I just mentioned that given a fundamental matrix at least you can get a pair of projection matrices in this form. But as we mentioned there should be a family of projection matrices pairs of projection matrices which will give you the same fundamental matrix and this family is shown here also. Once again, the results we will discuss the results, but not the derivation.

So, here we can see that these are the equivalent pair's projection matrices which will give you the same fundamental matrices or this is a family of fundamental matrix. So, what is the definition here. See v is any 3 vector and k is a scalar constant. So, the ke' is the same epipoles that you expect there and e'v^T it is a 3x3 matrix once again you can multiply with.

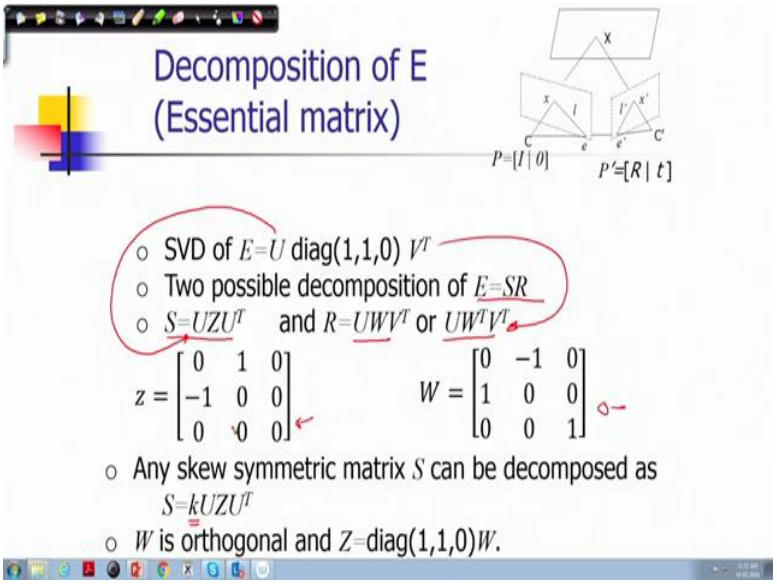
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For essential matrix, we can also derive the camera matrices as you can see that essential matrix is given in this form. So, once again this is a $[t \quad]_X$ is a skew symmetric matrix and R is a rotation matrix which is an orthonormal matrix. So, this is a kind of decomposition. So, given an essential matrix, if I get a decomposition of skew symmetric matrix and rotation matrix which is an orthogonal matrix; then I should be able to get at least a pairs of projection matrices given P = [I | 0] and P' = [R | t].

And the other fact we have already mentioned that two of its similar values are equal and the third one is 0. So, the matrix decomposition of E should give me the corresponding pairs of matrices.

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Now, there is a technique which ensures this decomposition. So, we have to perform singular value decomposition of essential matrix in this form, what is given here as you can see and there are two possible decomposition of essential matrix in the form of a skew symmetric matrix and rotational matrix; where you can see that it is a this E = SR and S is given in this from UZU^T;

So, U comes from the definition of the singular value decompositions and R is UWV^T or UW^TV^T. And you can find out the form of z and W.

$$z = \begin{matrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}$$

$$W = \begin{matrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}$$

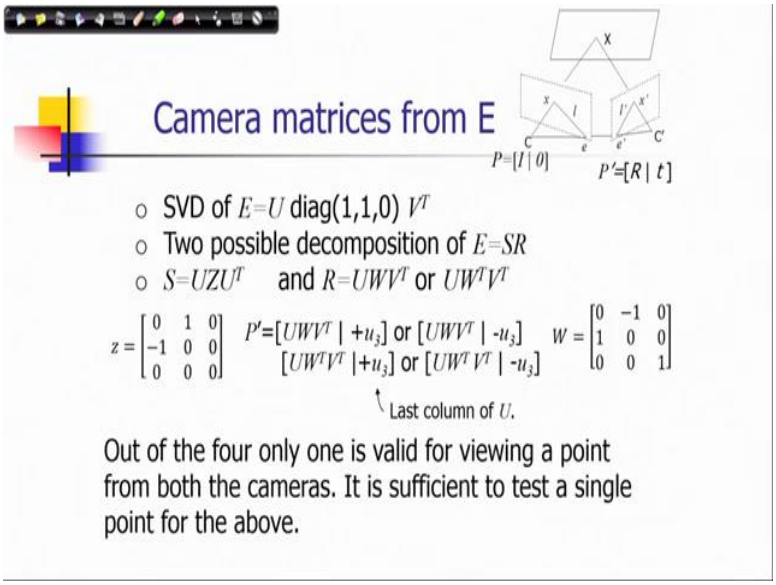
So, this is one particular solution. You need to perform the single value decomposition and then, z and W, they are well defined and you can get the corresponding matrices.

So, some more elaborations at any skew symmetric matrix S can be decomposed into this form

$$S = kUZU^T$$

you can show and even if we can multiply with a scale k as I mentioned that always for any fundamental matrix or essential matrix, you can multiply with a scale and you can show that W is an orthogonal, in the orthogonal form and Z=diag(1, 1, 0)W. So, there is a relationship between Z and W, we will see the motivation of why we are explaining this properties.

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So, by exploiting those properties, these are the possible configurations of the projection matrix of the second camera. So, first cameras projection matrix is already taken as [I | 0] in the canonical form. So, these are the possible configurations as you can see there are four such options are there

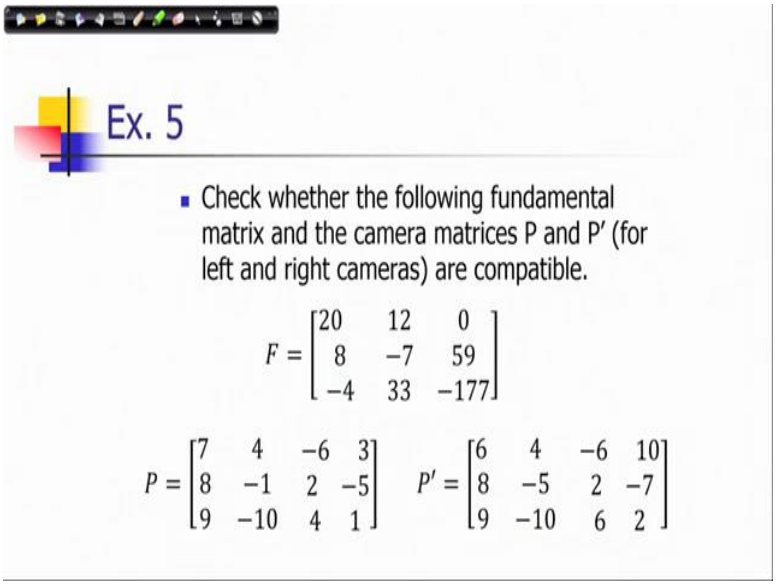
$$P' = [UWV^T|u_3] \; or \; [UWV^T|-u_3] \; or \; [UW^TV^T|u_3] \; or \; [UW^TV^T|-u_3]$$

You note the definition of u_3 here which is the last column of the matrix U which is given by this single value decomposition which means u_3 corresponds to the singular value 0 of U.

And but out of this four, only one is valid for viewing a point from both the cameras. We have already discussed that given a camera matrix how you can decide which point is in front of the camera. So, one of this projection matrices will have the same frontal

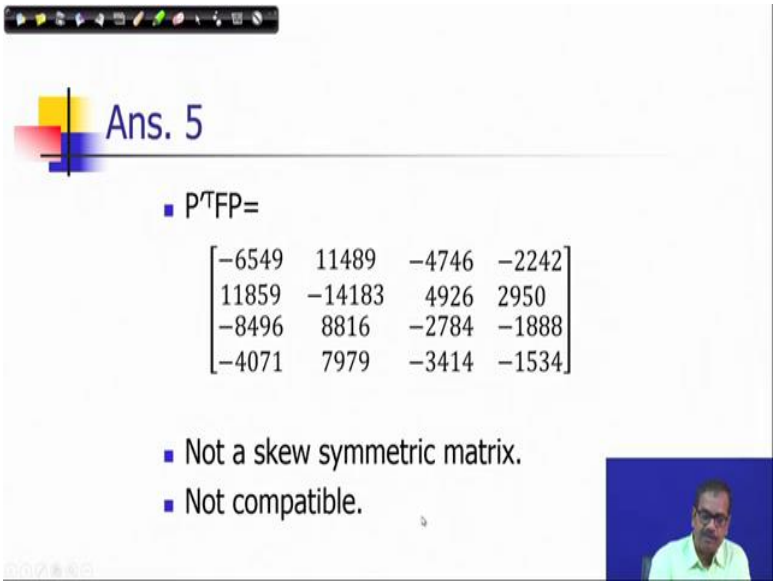
directions, same directions with [I | 0] and you should consider that matrix which satisfies. So, this is one form of solution of projection matrices from essential matrix.

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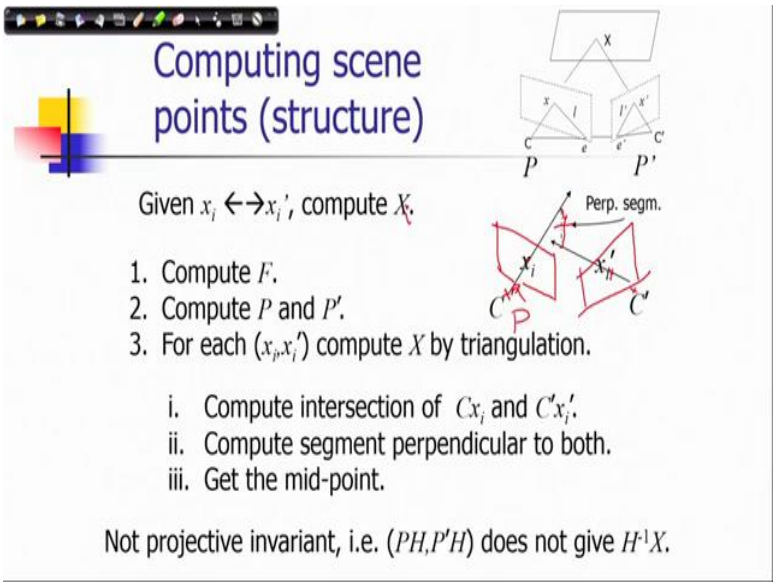
So, let us consider an example here using those properties. Now this problem says that you have been given a fundamental matrix and also the projection matrices P and P' of this stereo vision system. Now you have to check whether they are compatible or not. We discuss this particular property we have to exploit the property that P^T or $P'^T FP$, it should be skew symmetric.

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So, we need to compute this particular element, particular entity and if I perform matrix multiplications, I will get this matrix. And you can observe that this is not a skew symmetric matrix. So, which can readily tells us this fundamental matrix is not compatible with this cameras which is not compatible.

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We will now consider another computational problem that given a pairs of corresponding points, a set of pairs of corresponding points; how can you compute the scene points. We sometimes say for the whole and symbol as the structure of an object because if you know the three dimensional coordinates of the object points, we consider we have recovered the structure. So, essentially, we would like to compute the scene point ith scene point I can say also x^i scene point whose corresponding points are given by $x^{i'}$

So, one of the strategy could be that I can easily compute fundamental matrix F. Then, I need to find out P and P'. So, which may not be unique, but at least it gives the possible structure and then, I may later on do post processing to get proper combination of P and P' to get a structure. But right now, let us consider only that whatever P and P' we get, we need to compute the corresponding scene points.

Then we can apply the triangulation. So, I will explain what this method is by which you can get the scene points. So, triangulation method, it says that if I have the scene point x_i. Say you have an image plane corresponding to this camera, whose centre is here C and say this is your image point x_i. So, I can always form given its projection matrix P, I can form

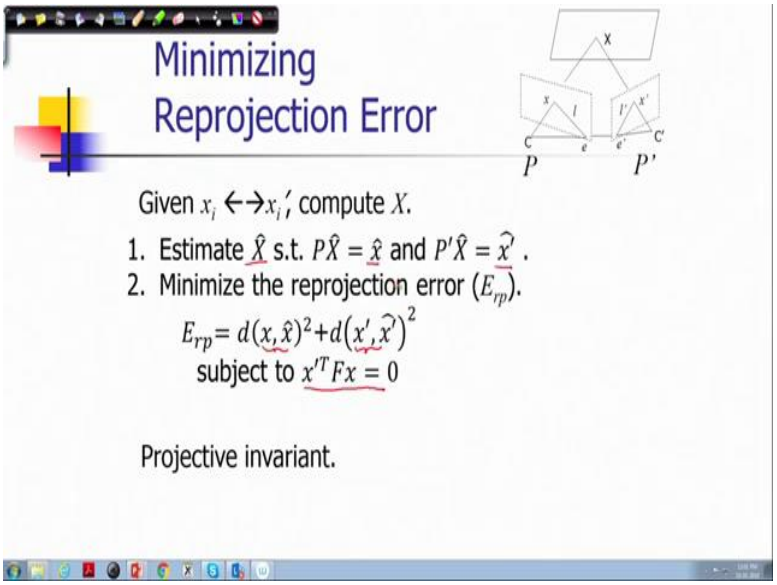
the back projection rate. So, I can form the equation of the projection ray, we have discussed earlier.

Similarly, for the other camera I know its camera centre; centre of the camera and its scene point I can form the other projection ray which means I would get the equations of straight lines you need three-dimensional space. So, now, the ideally, they should have intersected to my scene point; but there would be errors due to observation, errors due to computation. So, they may not exactly intersect. So, what I should consider, I should consider the closest point which should have been an intersecting point.

So, consider a perpendicular projection from this line to the other line. So, that is a linear segment and take the middle of that line segment and we will consider that is my scene point that is a solution. So, this is what we will compute intersection of Cx_i and $C'x_i$ front and for that we need to compute the segment perpendicular to both and get the midpoint.

So, we should note that this computation is not projective invariant because as we mentioned that camera matrices could have been (PH, P'H). Of course, if it is (PH, P'H), you would have got $H^{-1}x$. But the thing is that the way you are doing you would not get $H^{-1}x$ if you take also (PH, P'H). So, that compatibility is not there. It would be close to that, but it is not theoretically it will not ensure that you will get $H^{-1}x$.

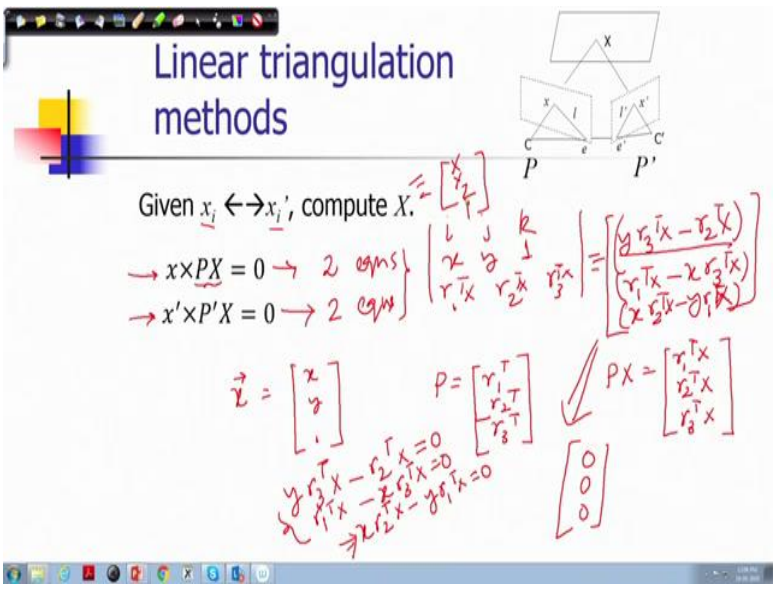
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Another method could be that we can estimate the scene point by minimizing certain objective function by considering the projection of those scene points applying the projection matrices. So, in this case you can see that what you can do is start with an initial estimate and then, observe what is your image point. Now, those image points should be close to your observed image point.

Then, this is the image point you obtain from computation and these are the observed one. So, if your estimate is good enough, this error should be very small and they should also satisfy this constraint. So, once again this is a constraint based optimization problem it is a non-linear optimization. There are various non-linear optimization techniques by which you can solve and you can define your estimates and this method is particularly projective invariant. That means if you take (PH, P'H), using this method the solutions will also satisfy $H^{-1}x$ and $H^{-1}x'$.

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And the linear triangulation methods we have discussed I think this is [FL]. So, in this method we will be using the algebraic manipulations. In the previous case of triangulation, we considered only geometric method of computing the intersections of two lines. But here we will formulate this problem as a solution of a set of linear equations. So, we will form a set of linear equations, as we did in previous cases also you will find out similarity; we will find the similarities of this formulations.

Since we have two pair of points x_i and x_i', we can consider that it forms an equation say Since PX is a same PX gives me the same point, but once again since it is a projective element its it could be scaled version of X, but the direction will remain same as a vector. So, if I take the cross product with X that should be equal to 0.

So, this will give me a set of linear equations to for which satisfies this constraint. Similarly, x'xP'X =0 that would give me another set up equations. So, if I expand this particular form, I can write it in this way say

$$P = \begin{matrix} r_1^T \\ r_2^T \\ r_3^T \end{matrix} \; and \; PX = \begin{matrix} r_1^T X \\ r_2^T X \\ r_3^T X \end{matrix}$$

See you see that this is a 3 vector and similarly, x also a point where we can represent in this form so

$$\vec{x} = \begin{matrix} x \\ y \\ 1 \end{matrix}$$

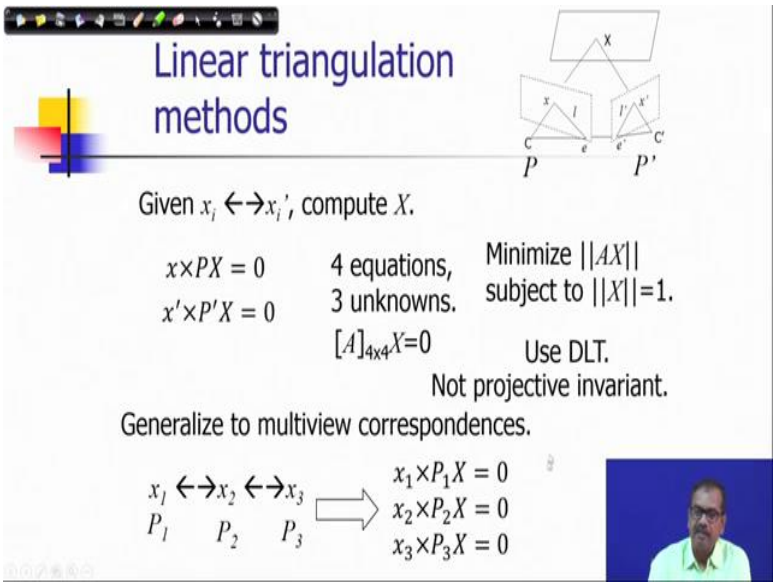
So, I have to take the cross product of these vectors. So, I will form this equation (x, y, 1), then $r^{1T}x$, $r^{2T}x$ and $r^{3T}x$. This is a cross product and there as you can if I expand these cross product once again. So, you will find that

$$yr_3^T X - r_2^T X = 0, \qquad r_1^T X - X r_3^T X = 0 \; and \; X r_2^T X - y r_1^T X = 0$$

So, we will find actually the third equations can be derived as a linear combination of these two.

You multiply with x and multiply with y and then, if you add them you will get the third equation. So, which means there are only two independent equation. So, it will give you only 2 equations. Similarly, this constraint will give you another two equations. So, given a pair of corresponding point you have 4 equations and how many unknowns you have for x? You have only 3 unknowns because you know these are the world coordinate points x, y, z and there is the scale dimension. So, there are 4 equations and 3 unknowns. So, once again you can apply the least square error estimate technique that we did.

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So, just to summarize no this particular computation, we can consider that there would be 4 equations, 3 unknowns as we mentioned and this can be expressed in this form set of linear equations. So, once again the same it is a set of homogeneous equations. So, we have to minimize the norm AX subject to norm X to be equal to 1 that is the constraint we will impose.

And you can apply this direct linear transform that we discussed earlier also for estimation of projective transformation matrix or homography matrix and this is not a projective invariant. We can generalize this particular construct that is the power of this technique. Because, if you have 3-point correspondence; that means, you have a three-camera system say P_1 P_2 P_3. So, each one will give me the similar form of equations. So, each one means each one will give me 2 equations. So, there are 6 unknowns; sorry 6 equations, but there are again 3 unknowns. So, you can solve using the same technique.

So, I think with this let me stop here we will continue this discussion of structure recovery in our next lecture.

Thank you very much.

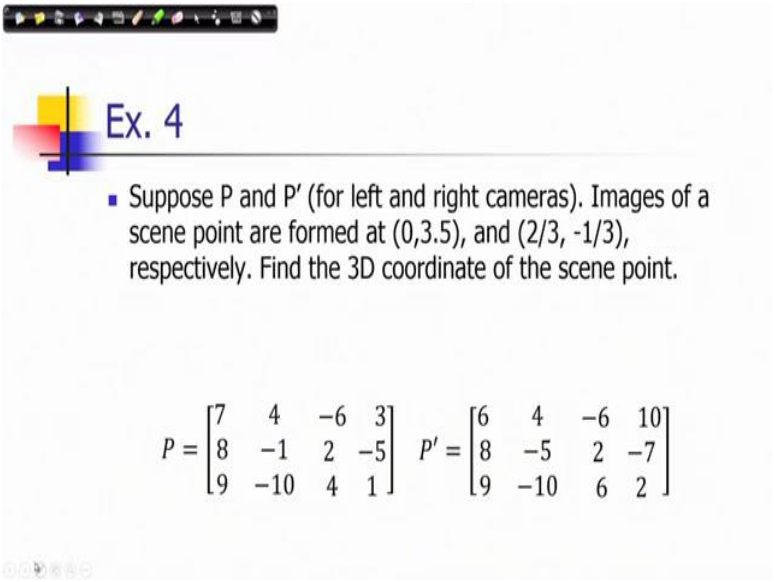
Keywords: Camera matrix, Projection matrix, fundamental matrix, triangulation.

Computer Vision
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Lecture – 22
Stereo Geometry Part – VII

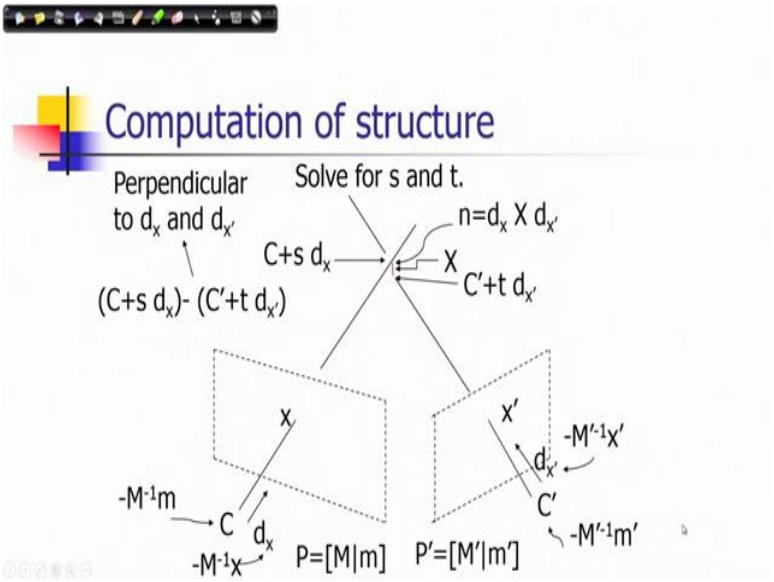
We are discussing about recovering structure from the set of corresponding pairs of points of a stereo system. And we have discussed how there are different approaches for computing that; computing a scene point given a pair of corresponding point. So, just to illustrate this process we will be discussing a problem for computing the scene point.

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So, consider this particular example that you have two camera matrices P and P' for left and right cameras. And its images of a scene point are formed at say (0, 3.5) and (2/3, -1/3). So, we have to compute the 3D coordinates of the scene point. You can see the projection matrices of the two cameras P and P' they are also given here.

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So, we will discuss the solution. So, let us consider how we are going to compute this particularly to get the scene point. So you have projection matrices P and P';

$$P = [M|m] \;\; P' = [M'|m']$$

And we have shown that two pairs of corresponding points as x and x'. So, we will be applying the geometry approach of triangulation; that means, we will form the rays back projection rays for these two cameras and then try to compute its intersection. And due to the noisy observation, you will not exactly get that intersection of this it is expected they should intersect, but you will not get exact intersection. So, what you need to find out that find out a very close estimate that intersecting point. So, you should consider a perpendicular segment for both the lines and then get the midpoint that was the solution.

So, let us see how this computation proceeds. So, C is the centre of the first camera, C' is the center of the second camera and we can compute the direction cosine of the projection ray; given the image point x. Similarly, direction cosine of the projection ray of the second camera; given the image point x'; so, this is what. So, $-M^{-1}m$ that would be the center C, then $-M'^{-1}m'$; that would be the center of the second camera, $M^{-1}x$ is a direction cosine of the projection ray of the first camera and $M'^{-1}x'$ is a direction cosine of the second camera.

So with this computations you get to know all the three dimensional constructs of the straight lines because the points passing through the straight line and its direction cosines.

Now, you apply the three dimensional coordinate geometry to solve the perpendicular segment and its meet point; so there lies the solution. So, one of the property of this perpendicular segment that its direction cosine should be obtained by the cross product of dx and dx' because it is perpendicular to that.

And if I assume; a point lying on the projection ray passing through the image point x and C; as C plus s d x; so s is some it is a parametric representation of that straight line; any point can be; if a s varies; s is a any positive value. Similarly, C prime plus t d x prime is the another point. So, what we need to find out exactly that which s and t will satisfy this constraint that the line joining them; line joining those two points given by this values of s and t will give me a perpendicular directions of both the rays; so that we translate. So, it is perpendicular to this and then you know we need to solve for s and t applying this constraint.

$$C = -M^{-1}m$$

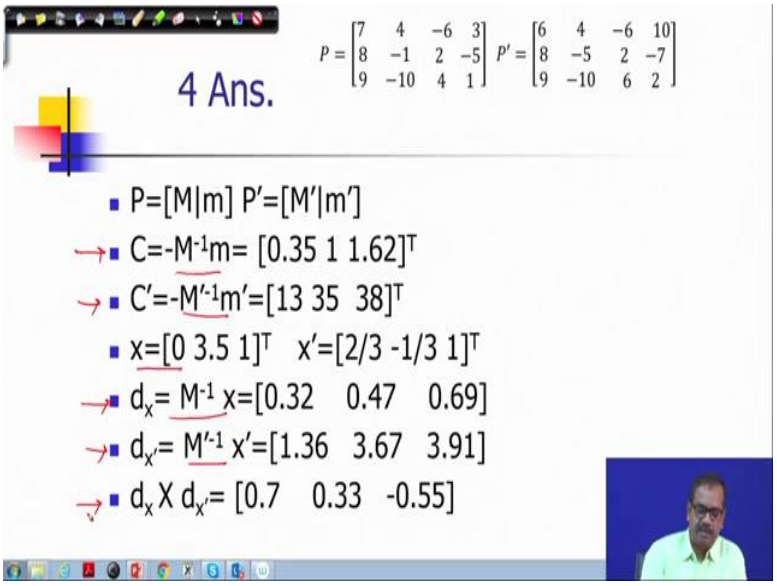
$$C' = -M'^{-1}m'$$

$$d_x = M^{-1}x$$

$$d_{x'} = M'^{-1}x'$$

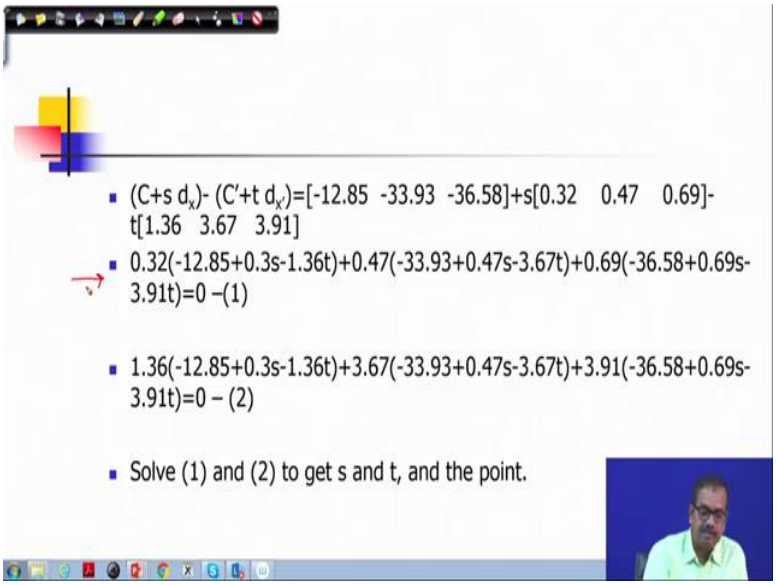
$$direction\ cosine = d_x X d_{x'}$$

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So, these are computational steps what we discussed and here is the summary of all those results. As you can see that we have computed the first centers, this is given by applying this relations. Then you have completed the direction cosines by applying this particular facts. Then this is the direction of the perpendicular segment which is perpendicular to both of these rays.

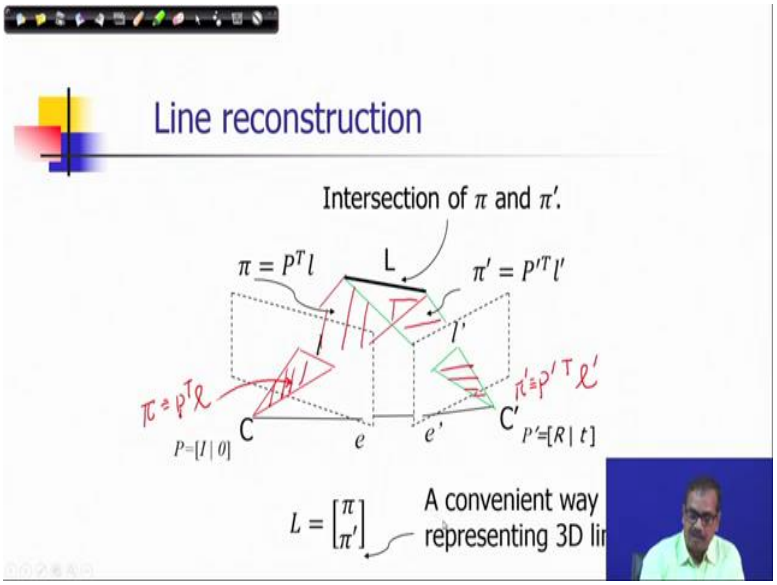
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And then we are trying to solve for s and t by applying the constraint that the line forms by this; so the line forms by them this segment it is perpendicular to both the directions of dx and dy'.

So, you get actually two equations; so if you note the direction cosine of dx=[0.32, 0.47, 0.69]. Then in this slide you can see that you have taken the dot product of that direction with this particular expression that should be equal to 0. Similarly from the direction cosine of dx'; it should be orthogonal to this direction. So, the dot product of between these two vectors should be equal to 0. So, this gives me two equations is that a linear equations in terms of s and t; so if you solve and you can get s and t and the point.

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So, this is how we can solve this problem and we can get the structure. Let us consider the other problem of line reconstruction. So, you consider here a particular situation, where you have been given a straight line L and you can get its two images which will be again straight lines in the first image plane and in the second image plane.

Incidentally, as you can see that there will be also an epipolar lines those corresponding epipolar lines. So, this is an image of 3 dimensional line in the first image plane; similarly the image of the same line in the second image plane; both are given by L and L'. So, what we can do here that given this line and given this projection matrix P; we know that equation of this plane is $P^T l$; so this is the equation of this plane.

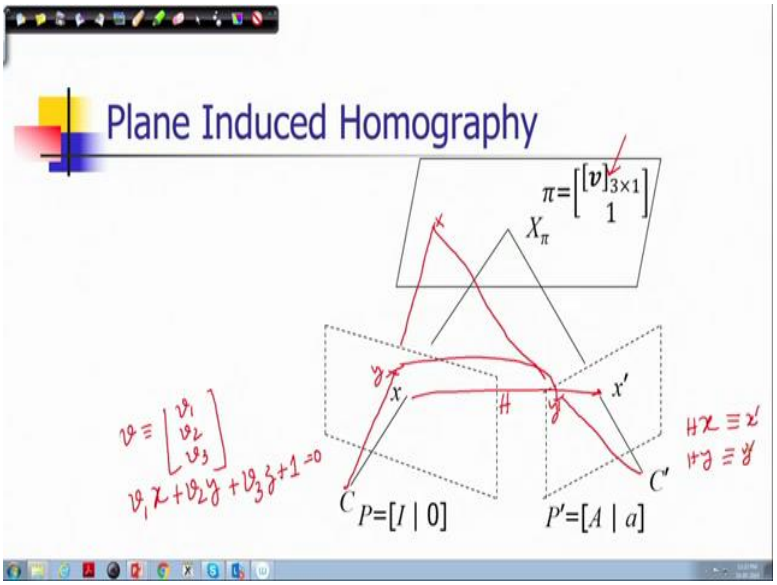
$$\pi = P^T l$$

$$\pi' = P'^T l'$$

$$L = \begin{bmatrix} \pi \\ \pi' \end{bmatrix}$$

Similarly, equation of this plane is $P'^T l'$. So, now you have these two plane one is say π another is say π'. So, when two planes intersect; they forms a line and that is the solution there is a three dimensional line and that is exactly you will get if you compute this intersection; so this is what we I just mentioned. So, finally, your solution is in this form; so this is one convenient way of representing a three dimensional line.

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Now, we will be discussing a very important property of a stereo system; an important feature of a stereo system. And you can show that how the projective transformations among the image points they exist in a stereo geometry. So, the homography or the projective transformation are established among the pair of corresponding points for which the scene points lie on a particular plane. So, we call this kind of homography a plane induced homography.

So, take this particular scenario which is been shown in this diagram that you have once again two cameras of projection matrices P and P'. So, P is given by [I|0] that is the projection matrix for the first camera and P' given in this representation [A|a];

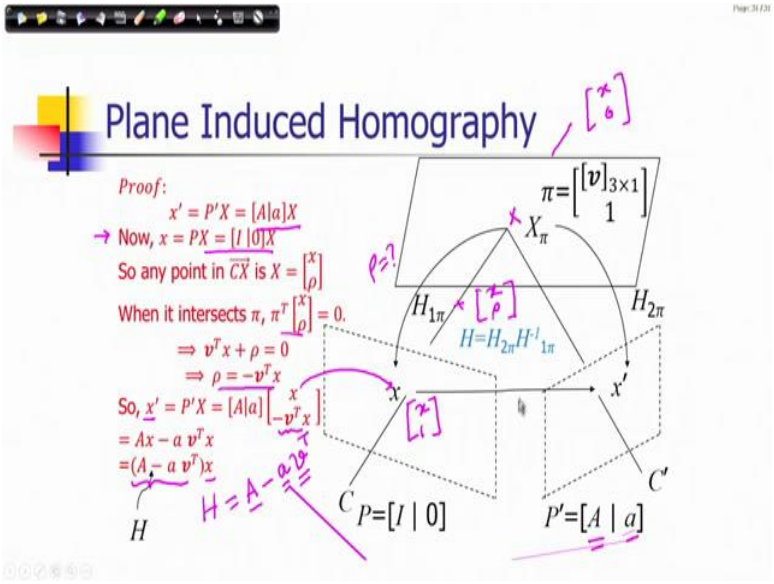
And we have a plane which is given by this algebraic representation as you can see this is an equation of a plane. So, representation of a plane in a homogeneous coordinate system of a three dimensional projective space; so this is the 3 vector which is giving you the directions cosines of the plane and your scale value is kept as 1; which means the equation of the plane if v is given as a column vector

$$v = \begin{matrix} v_1 \\ [v_2] \\ v_3 \end{matrix}$$

$$v_1x + v_2y + v_3z + 1 = 0$$

Consider a point on that plane which is given here x pi and its corresponding image points x and x prime. So, we can show that there exists a homography between these two points. So, you can take another point for example, and also you can form its say images say this is y and say this is y prime; so this is the corresponding point. So, there exist homography H such that Hx will give me x prime and Hy will give me y prime and this homography is called plane induced homography; we can show this things. So, let us check the derivations, clean this particular diagram.

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So, we can see that; we have already discussed earlier also that points in a plane and its corresponding image points in a single vis[ion]- camera also they are related by the homography. Because points in a plane can be represented by a three vector by two independent parameters and they are related by a homography; consider that homography is H 1 pi. Similarly, you can have an homography H 2 pi; so from there you can get a homography which is given by this H 2 pi; H 1 pi inverse.

This plane induced homography is can be related with the parameters of the plane which is inducing this homography and also the parameters of the projection matrices. So, here we have shown; we have shown a proof that how it is related the; the basic theorem is that this homography is given by this. That means, plane induced can be expressed in this form where A is the 3 cross 3 sub matrix of the projection matrix of the second camera or P prime. And this vector small a is the fourth column vector; fourth column vector of this

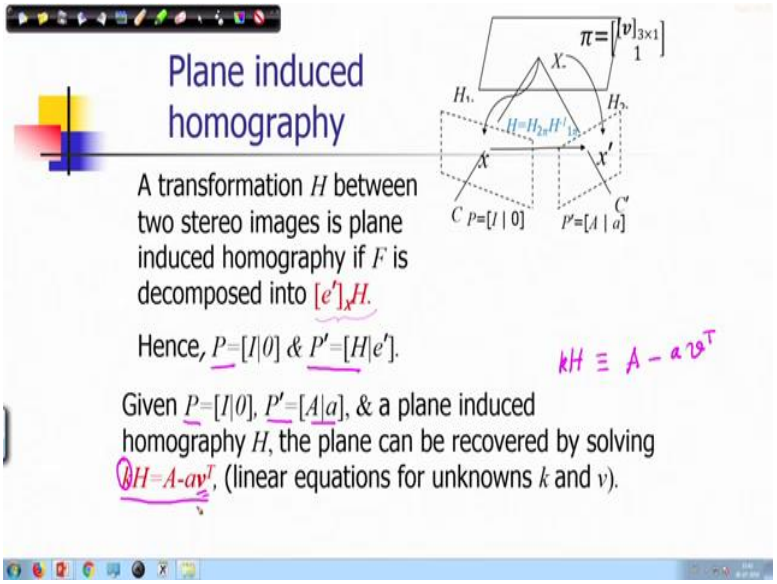
projection matrix and v as you know this is the direction ratios of the normal of the plane pi.

So, this has been shown here how it be related like this. So, let me just provide you the outline of this proof as it is given here. Now, you can see that x prime which is the image of the scene point x is given by this P prime X; which is can be expressed in this form whereas, x is just PX; which can be expressed in this form. So, now, any point in this ray can be considered as x rho; as you know that the in the homogenius coordinate system, the scale factor provides you the corresponding you know points and the interpretations with respect to three dimensional space real space is that; it should could be any point in that projected ray connecting the image point x.

You should note that at rho equal to 1; you have the image point. So, when rho is 1; this is represented in this form and when at infinity, this would be represented in this form x 0. So, exactly at what value of rho you are getting this value that can be determined by considering the point containment relationships on the plane. So, if I take pi transpose x rho; this vector that should be equal to 0 and value of rho is given by minus v transpose x and that now we will be using in the expression of x prime.

So, x is related with this capital X that is the scene point is related with the image point x and also the corresponding scale factor which has been computed using this particular fact of point containment in plane. So, x and scale factor is minus v transpose x; so if I do the corresponding submatrix multiplication, I will get it as Ax minus a v transpose x and no ah; so this can be written in this form. So, as you see x prime is related with x; by this linear transformation and which is nothing but we call it a projective transformation and which is what is homography H. So, you will get in this way this is the direct relationships.

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So, using this particular fact we can say that a transformation H between two stereo images is plane induced homography; if you can say in the other way that if fundamental matrix can be decomposed into this form; when e prime e prime cross H; e prime is a epipolar plane epipole epipoles right epipoles. And as you know that the cross product of right epipole with the matrix that is an operation matrix operation basically this e prime cross is a matrix equivalent matrix which is representing that operation; that itself will be we can you will give you F.

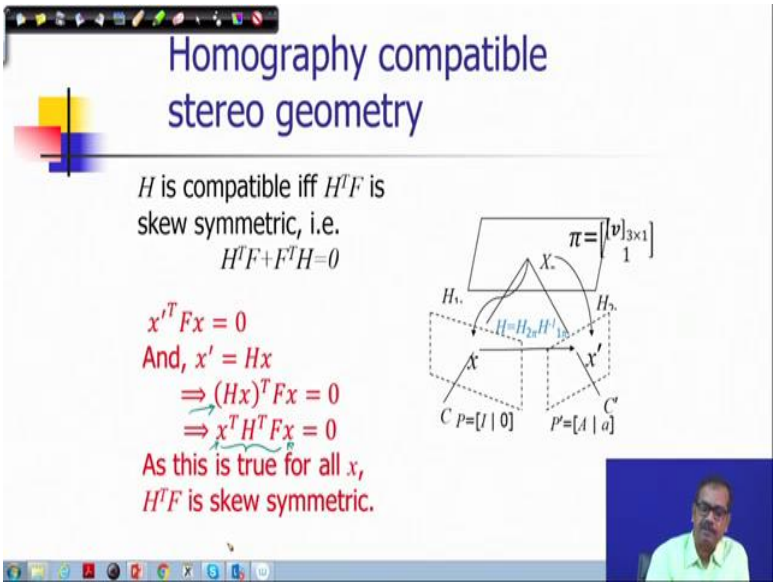
We have; earlier we have discussed that actually this homo[graphy]- it could be any homography and that is what it is represented here; previously we have taken only the homography at plane at infinity. So, even if I decompose F in this form of e prime cross H; if we decompose F in this form of e prime cross H, then P and P prime can be ah obtained; can be can be expressed in this form, it is a in the family of ah pairs of projection matrices this could be also one such option.

So, we can also say that if you have two projection matrices P which is in the canonical form of I 0 and P prime which is given in this form as we can see that a plane induced homography H. And then the plane can be recovered by solving this equations because we know that homography would be ah; we have seen that homography is given by A minus a v transpose in the previous case. But naturally this equality we cannot establish unless

we know what is the; exact equality can be determined can; can be put when we know exactly what at what scale this equality holds.

So, that is why we have to we write this equation in this form. So, what are the unknowns here? There you have the unknown k and unknown v; so there are you get a linear equations there will be no; no there in the H, there are 8 elements; 8 independent elements independent parameters. So, you can get linear equations for unknowns k and v and that you can solve using this homography. So, this is how you can get the equation of the plane.

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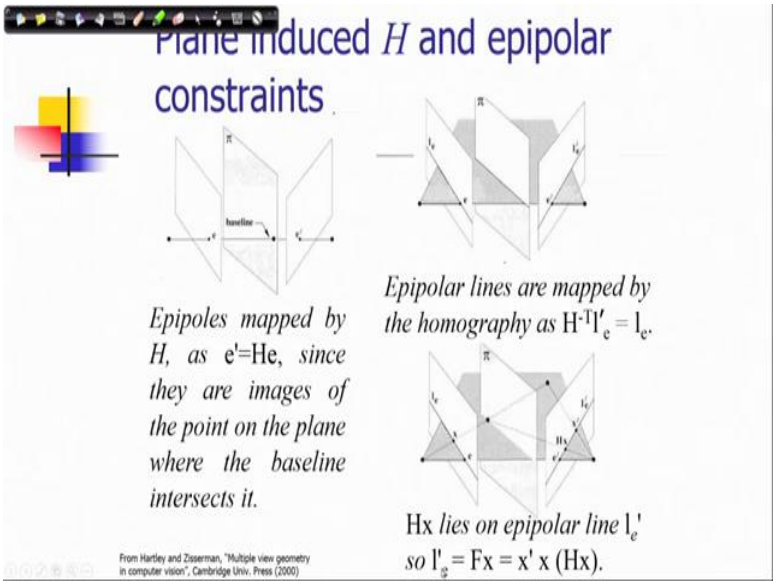


Another feature of the stereo geometry that this plane induced homography or you can find out whether it is compatible to a fundamental matrix or not. Like you have observed; we have discussed how fundamental compatibility of fundamental matrix will pairs a projectional matrices could be tested by observing, whether it is that there is a skew symmetric matrix property; P prime transpose F P should be skew symmetric. Similarly here also you can see that H transpose F should be skew symmetric. So, that is you know there is one of the interesting that is one of the feature of the skew symmetric matrix as the transpose of that matrix is the negation of the matrix. So, if we add the transpose you should get 0.

So, this can be shown in this form that is the proof that since x prime transpose F x equal 0 and x prime is H x. So, H x transpose F x equals 0 and this is true; once again this is true

for every x because see these are all same image points; so this is true for every x. So, H transpose F is a skew symmetric matrix from the property of a skew symmetric matrix.

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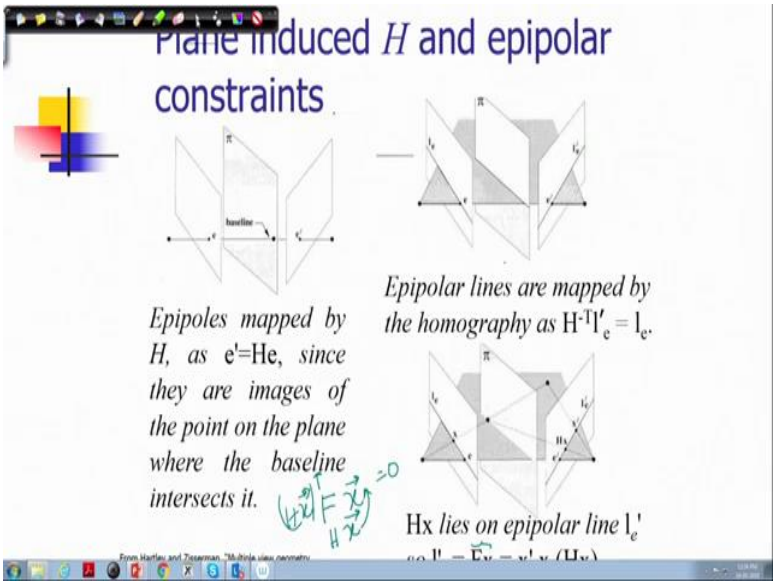


So, these are certain facts of the stereo geometry that once you have a plane induced homography; that means, in this case the corresponding pairs of points they are all formed from the scene points of a plane, so that is explained by that form. See you consider, a plane any plane you consider and in that plane if it is not parallel to the base line; there is an intersection with respect to that plane. So, once again if for that plane if you know there is a homography H; so e prime should be equal to He for that plane induced homography.

Similarly, we have mentioned that epipolar lines; corresponding epipolar lines which are lying on the same epipolar plane. And any particular line lying on the same epipolar plane they also form the plane induced homography. Because once again the plane which is inducing and points are lying in that plane and any intersection of that plane with respect to epipolar plane will satisfy this constraint. So, this is also satisfied; for any pair of epipolar lines this is satisfied.

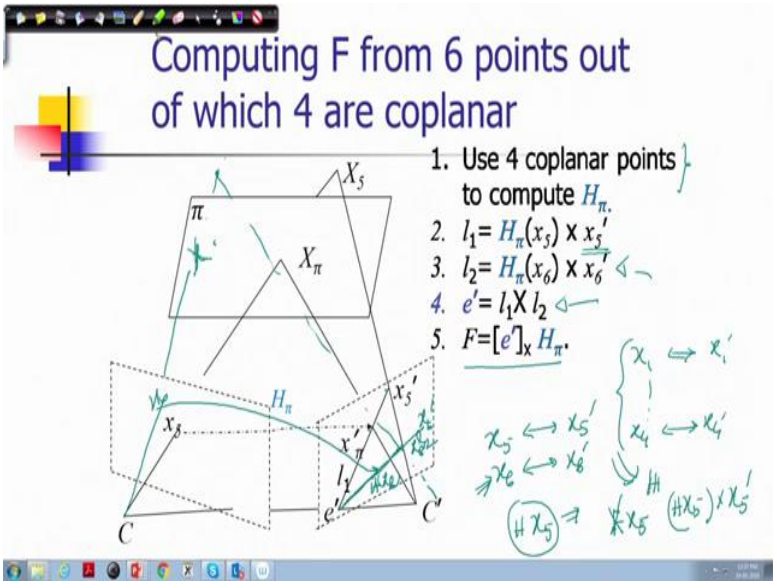
And H x lies on epipolar line of L e prime, there is another constraint it has been shown that here also you can see that if you have; you have a scene point if you have some scene point here this is the epipolar line. So, epipolar line of a corresponding point can be obtained by performing the multiplication with respect to x F x.

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And then; that means, I can obtain the epipolar line by multiplying x and if I perform the homography transformation on x; so this point will also lie here; that means, I can write H x transpose; F x that should be also equal to 0. So, that is the essence of this particular computation.

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So, we will be considering; we will be exploiting this facts for estimating the fundamental matrix under certain constraint scenarios because that would simplify the computations. Suppose, you would like to compute fundamental matrix from 6 points; out of which 4 are

coplanar. So, I can using that 4 coplanar points; we know that they establish a homography. So, their corresponding points; there is a relationship of homography among themselves.

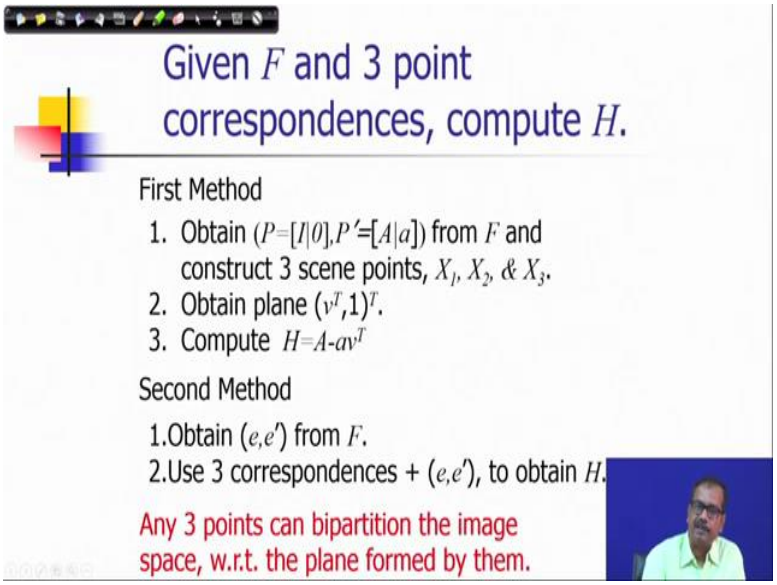
So, we can compute the homography matrix H by using those 4 points and then you can find out the epipoles by computing the epipolar lines. So; that means, you consider the actual corresponding points which is been observed for x 5. So, you have x 5 corresponding to x 5 prime, then x 6 corresponding to x 6 prime; also you had x 1 corresponds to x 1 prime to x 4 corresponds to x 4 prime. So, first you compute the prime induced homography among themselves, you require minimum 4 point correspondences.

So, now if I transform x 5 using this homography; the same point will lie in the epipolar line f x 5 or I should say that; the epipolar line will be formed because F is not known. So, epipolar line will be formed by H x 5 cross x 5 prime. So, that is what is shown x 5 prime and this is the plane induced homography point; so they will form an epipolar line.

In the same way, using the other point you form the second epipolar line. So, you will get another points see you get x 6 prime and it is suppose this is x 6, this is x 6 the; it may be somewhere else, it is not here. So, this point may be here and say this is x 6 prime and after homography transformation; this is H x 6. So, this will also form another epipolar line; so this two epipolar line l 1 and l 2; their intersection will be e prime; that is expressed in this from.

So, if I take the cross product of l 1 and l 2; I will get e prime. So, now fundamental matrix can be defined as e prime cross H pi that is fundamental matrix; so you can compute it. So, using the 6 point no fundamental matrix from 6 points; you can compute using the 6 points, you can compute fundamental matrix.

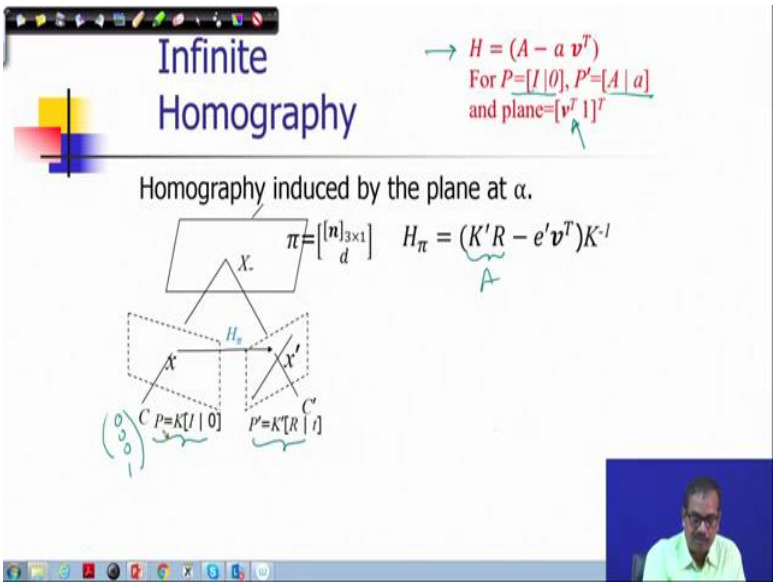
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Now, here the problem is the; in the reverse. You have been given a fundamental matrix and 3 point correspondences and then you need to compute H. So, the first method says you can fundamental matrix you can always; no estimate the camera matrices P and P prime using the method what we discussed. And then you can obtain the plane because you have 3 scene points and you solve for the 3 scene points; construct 3 scene points and obtain the plane and then from there you can compute the plane induced homography that is H equals A minus a v transpose given the parameter in this form.

The second method is that; that for that 3 point correspondences, you require another point correspondences from which you can get this homography because minimum 4 point correspondences are required. So, what are those, what is the pair of point correspondences that is the epipoles. So, from F; you can compute this epipoles; that means, you get its left 0 and right 0; those are the epipoles and they are also corresponding points. So, now you have 4 corresponding points from where you can compute homography. So, this is also another interesting feature that any 3 points can bipartition the image space with respect to the plane formed by them.

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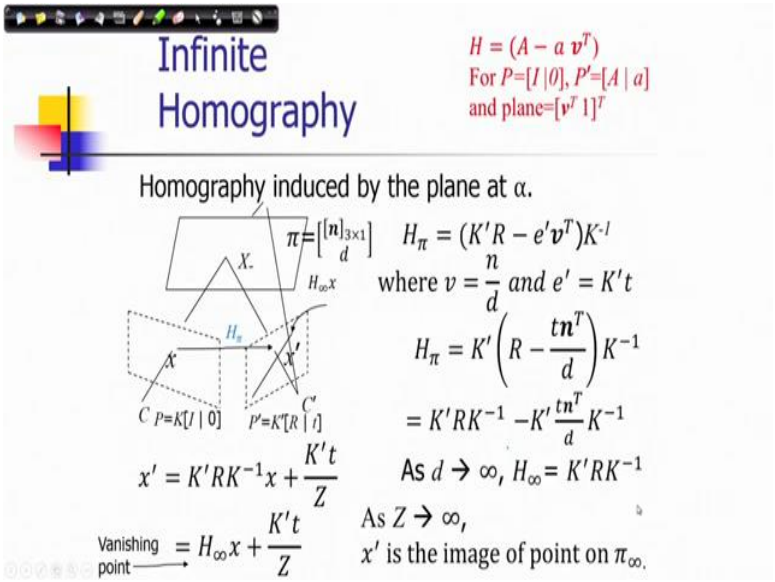


Infinite homography, we discussed earlier also and let us discuss it in the context of this plane induced homography. So, you consider any general plane induced homography which is the results are shown here at the top; here you can see that this is the general result of homography given your projection matrices in this form and also the plane three dimensional plane; the representation in this form.

So, in this case of course, this is given in the form of n d; so that we can relate it. So, you just if you just scale this normal vector by d that is equal to b; so let us see how this. So, now your plane in this tomography following this particular constrict could be written in this form. Now, here your projection matrices are given in this; so it is related with A; A is K prime R; so this is A. And your e prime is given by this know e prime is the image of these coordinate which is 0; centre is at since it is 0 center is at 0 0 at the origin. So, if I take the image here and we will get the corresponding; so it is t.

So, the epipole is image is at t right. So, this e prime we will find out e prime will be t and let us see what is K inverse? K inverse because this is I 0; so K inverse will come here. So, we can make it in a canonical form; if I transform the image coordinates by k inverse.

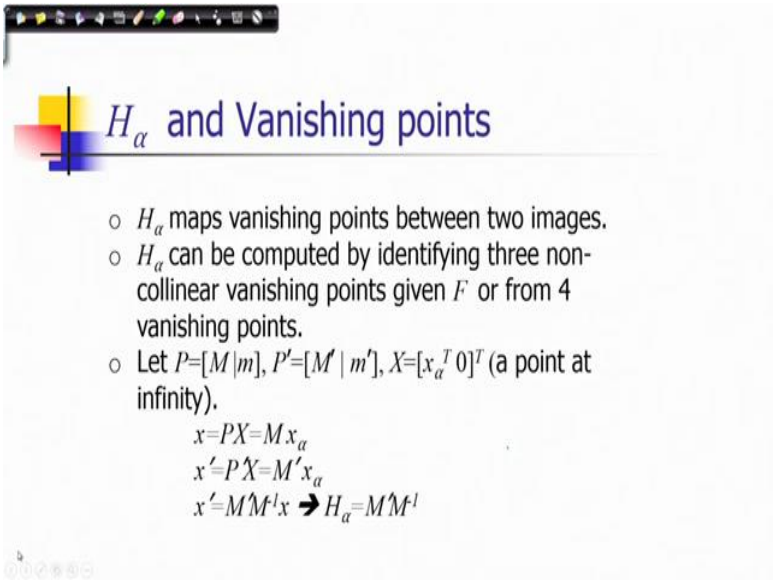
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So, these are the things; so v should be equal to n d and e prime is K prime t; e prime is K prime t from here; so that is what is e prime. And then H pi following this you know formulation can be written in this form. So, you note that v is given this n d; so it is t n transpose d. So, finally, it is K prime R K inverse minus this. So, now as d tends to infinity H infinity is K prime R K inverse.

This is a formu[la]- which we know discussed earlier also that is the homography at infinity can be expressed in terms of these camera parameters. And we discussed also earlier how a in general configuration how the corresponding pairs of points are related with this particular parameter; so this is what we get using infinite one. So, as Z tends to infinity; x prime is image of point on pi infinity. So, you this is; this shows you that that is a vanishing point over theepipolar line; as you are you are projecting the points at infinities.

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So, this is a summary between the line the homography of you know; homography at infinity at vanishing points H infinity. It maps vanishing points between two images and it can be computed by identifying three noncollinear vanishing points given F or from 4 vanishing points. So, we discussed earlier also that given three points and fundamental map fix; you can compute the homography.

So, if you have vanishing points ah and corresponding vanishing points; you can compute H infinity in that case or you can get 4 vanishing pair of vanishing points; you get again compute H infinity. And if your camera mattresses are given in this form P and P prime and any point at infinity is given in this form; we discussed earlier also this shows at H infinity is M prime M inverse.

This relationship we discussed while solving a problem itself that given a general camera representation, how homography at infinity or plane in this tomography when the plane at infinity is inducing plane is expressed in this from or H infinity is M prime M inverse. So, with this let me stop this lecture at this point we will be continue our discussion on stereo geometry in also subsequent lectures.

Thank you very much.

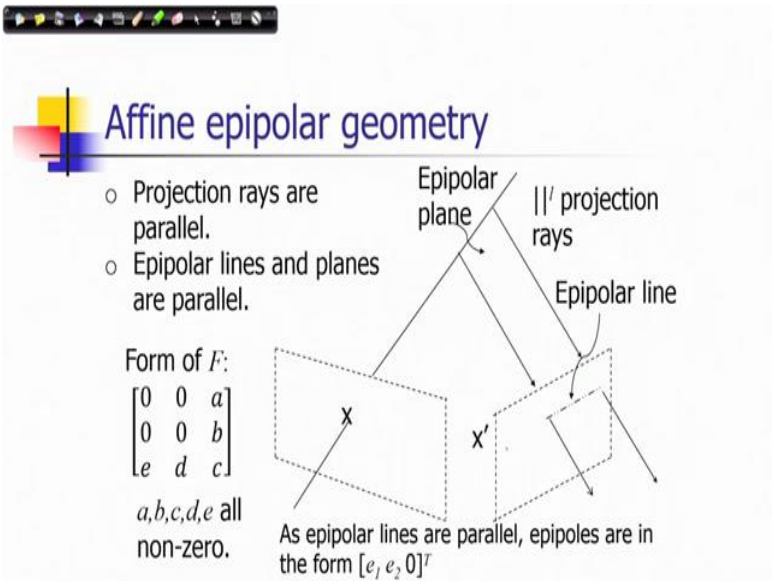
Keywords: Plane induced homography, line reconstruction, epipolar constraints, infinite homography

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Lecture – 23
Stereo Geometry Part - VIII

We will discuss special case of epipolar geometry, under the same topic of Stereo Geometry and that is affine epipolar geometry. So, we discussed earlier also how an affine camera differs from a general projective camera. In an affine camera your centre of projections or camera centre that lies at infinity which means that the projection rays, they are all parallel to sudden directions and the basic projections basic imaging mechanism it takes place through that parallel projection.

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So, let us consider a stereo system which is formed by two affine cameras. You can see here in this diagram that there are two such cameras and there are direction, so it is a parallel projection ray which is forming images in the second camera. Similarly, in this direction ray for the first camera. And the camera centers this is lying at the infinity, this parallel projection, so the intersection of this projection rays will be at infinity which will define the camera centre, effectively that is the direction of these rays itself.

So, how the epipolar lines are found? Epipolar lines are all parallel lines. Here we can see that for the same projection ray, if I take the projection of the same points lying in this

projection ray then all those image points will form an epipolar line. And since all the rays are parallel here all the parallel rays in the first camera, epipolar lines also should be parallel.

And the form of fundamental matrix particularly when you have this kind of geometry it get simplified you can see the structure here, you have only 5 non-zero elements.

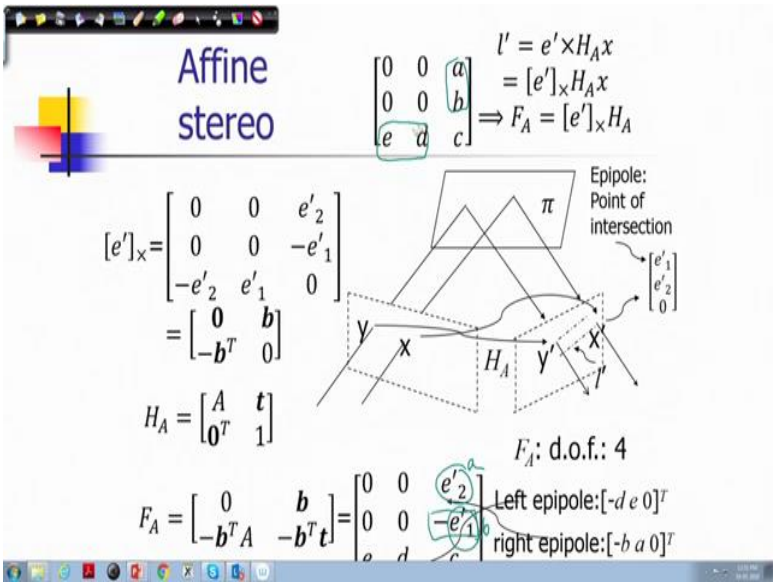
$$[e']_x = \begin{bmatrix} 0 & 0 & e'_2 \\ 0 & 0 & -e'_1 \\ -e'_2 & e'_1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & b \\ -b^T & 0 \end{bmatrix}$$

See all the 4 elements in this part they are 0, and the other elements are non-zero. And so there are only know 5 parameters, but since once again it is their scale is involved in this representation, so there are 4 independent parameters. So, here projection rays are parallel and epipolar lines and planes are also parallel. That is the characteristics of an affine epipolar geometry.

And the other thing is that as epipolar lines are parallel, epipoles are in the form of e1, e2. So, we know that because the intersection of epipolar lines that would be also at the point at infinity which means the value at the third dimension representing scale should be 0.

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So, let us again elaborate this particular property. Consider this affine stereo x is corresponding to x‘, y is corresponding to y’ and they are existing homography between them when all these points are lying on a particular plane pi. So, that is the plane induced homography.

$$H_A = \begin{bmatrix} A & t \\ 0^T & 1 \end{bmatrix}$$

And so epipole a point of intersection is represented as e1’, e2’0 we have seen that this is the structure. And, following the same construct the epipolar line is given by e‘XHAx because this is the homographic transformation of y to y’ which will lie also on the epipolar line and epipole also will lie on the epipolar line. So, e’XHx equal to 0, so FA should be e’XHx. And since epipoles have these construct, so all those 4X4 sub matrix of the affine fundamental matrix in the upper side that becomes 0, because of these structure.

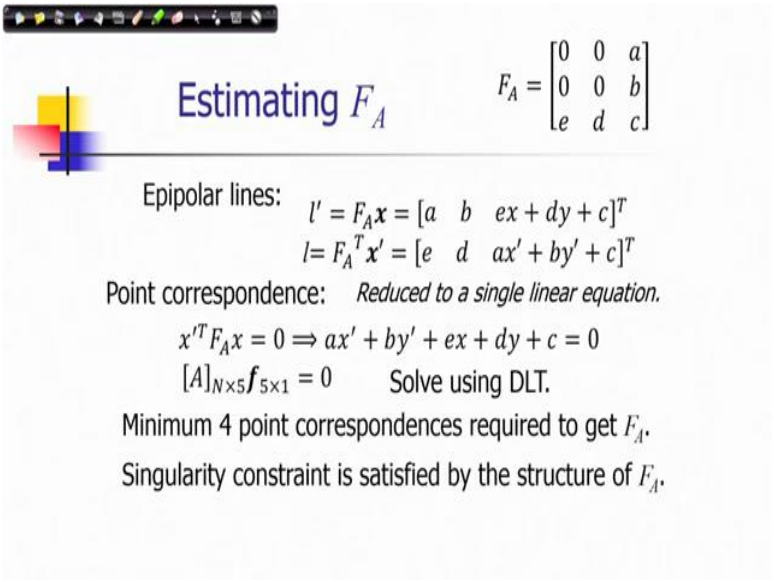
$$l' = e'XH_Ax$$

$$= [e']_xH_Ax$$

$$F_A = [e']_xH_A$$

So, particularly you notice that the right epipoles they these are given by this particular column. So, the values of the right epipoles it has occurred in this fashion. So, e2’ comes here and e1’ comes here so. Similarly, we will see also the values of the left epipoles will be at the row [-d, e, 0]. Right epipole is given by [-b, a, 0]. So, from the structure of affine fundamental matrix epipoles are very easily determined that we can see.

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$$F_A = \begin{matrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{matrix}$$

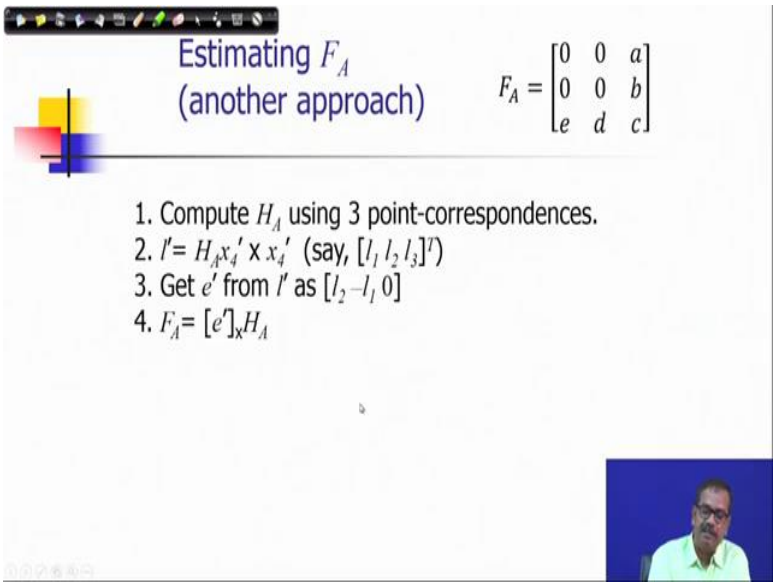
$$l' = F_A x = [a\ b\ ex + dy + c]^T$$

$$l = F_A^T x' = [e\ d\ ax' + by' + c]^T$$

For estimating this affine fundamental matrix we have to use the corresponding epipolar lines. So, this equation becomes a in a in a much more simpler form and you can represent the set of linear equations again in the matrix form in this way and you can solve using direct linear transform.

Since, there are 4 independent parameters you require only 4 point correspondences to get FA and from the structure itself singularity is satisfied because of the structure that all the upper sub matrix 2 x 2 sub are all of them are 0, so if you take the determinant you can check that the determinant value itself is 0 in this channel structure. So, you do not have to perform any special operations to input singularity.

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So, the approach was using the set of correspondence points where you can find a linear equations exploiting this constraints of epipolar geometry, and you have seen how it can solve those problems and solve those equations which becomes a simpler for the case of affine fundamental matrix estimation.

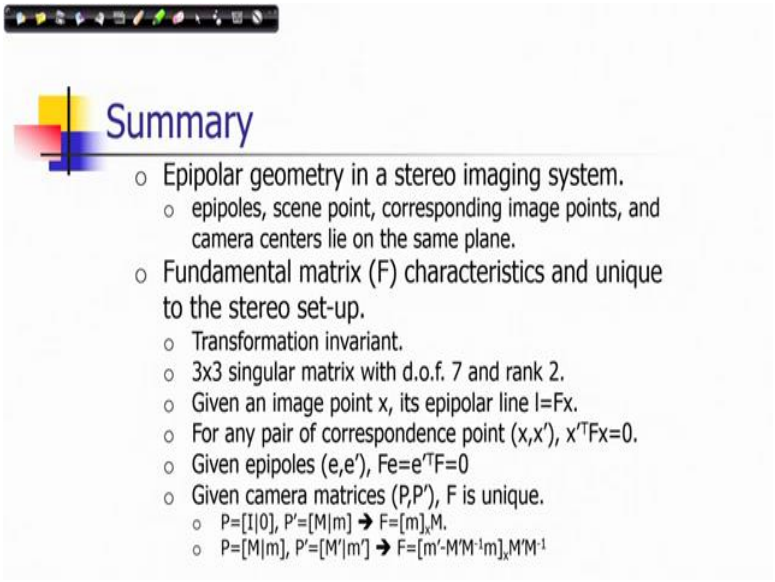
Now, we can use another approach for estimating the fundamental matrix. Here we can compute the first we can compute the homography induced by a plane by using 3 point correspondences. So, minimally we require 4 point correspondences. So, we have seen already this homography would be affine homography and for that 3 point correspondence is sufficient to compute it. So, we can do that and then we need to compute the epipolar line.

Now, in this case it is sufficient to compute the epipolar line l' because from there itself we can get the epipoles. The directions of epipolar line itself will give you that epipoles because that is what its direction cosine is given in this from [l2 -l1 0]. So, actually, epipoles is the point at infinity along that directions and this is direction of the line. So, in this way epipoles can be obtained.

$$l' = H_A x_4' X x_4'$$

$$F_A = [e']_x H_A$$

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So, we have come to the end of this particular topic of stereo geometry. So, let me summarize it is different features that we discussed. First we discussed epipolar geometry in a stereo imaging system and there we have seen that epipoles, same point corresponding image points and camera centers, all of them lie on the same plane. Then fundamental matrix characterize its stereo system and it is unique to the stereo setup it has various properties like transformation invariance is one of the properties. That means, transformation of image points will give me back the fundamental matrix. You can use the transform points to estimate it.

Then it is a 3X3 singular matrix with degree of freedom 7, that is the number of independent parameter is 7 and since it is singular its determinant has to be 0, rank has to less than 3 and in this case for a fundamental matrix its rank will be 2. So, it is function that it transforms an image point into its epipolar line in the second camera plane, second image plane. So, and this is the relationship that you have to multiply fundamental matrix with image point in the homogeneous coordinate representation, then you will get a line in the homogeneous coordinate representation in the image plane.

And for any pair of corresponding points you get this relationships

$$x'^{T} F x = 0$$

If we get epipoles e' then this is the relation

$$Fe = e'^{T}F = 0$$

So, they are the zeros of fundamental matrix. So, epipoles are the 0s of the fundamental matrix. e is a epipole in the reference camera plane, so it is right zero of F and e' is the epipole in the second camera plane and it is a left zero of the fundamental matrix. On the other hand, if I change the convention of reference camera that show up them then F transpose will be the fundamental matrix and all these results are consistent with that representation.

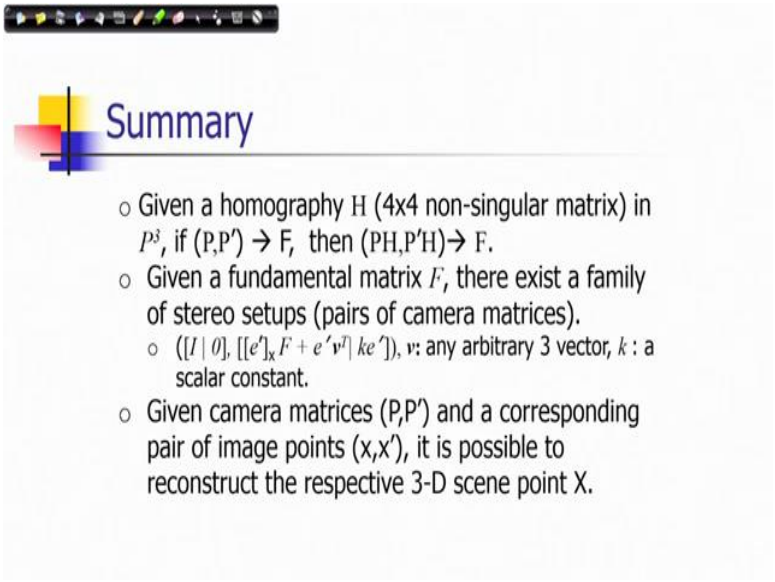
So, given camera matrices P and P' and F is unique that we discussed. And particularly these computations we should note that if we have the projection matrices in this from, that projection matrix in a canonical form of [I|0] and P' as [M|m], this is the representation. Then fundamental matrix is given as mXM. So, we discuss the definitions of these notations in our lecture.

Or, you consider a very generic representation I have given you the expression, I did not really expand or discuss in my lectures, but please go through it both note this expression. So, what we will find that from P and P' we have computed the epipole and which is given by this

$$F = [m' - M'M^{-1}m]_x M'M^{-1}$$

So, this is now the epipole is given. That means, you have to take the image of the centre of the first camera. So, image of the centre of the first camera is given in this form. And then the himography at infinity is given $M'M^{-1}$. So, that is how we get this fundamental matrix.

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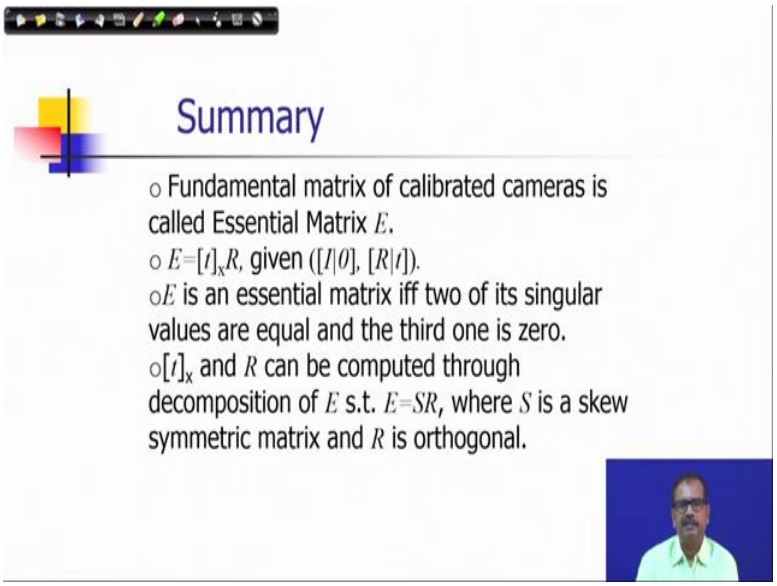


Then we discussed that how projection matrices could be related with fundamental matrix though a pair of projection matrix give uniquely one fundamental matrix, but a fundamental matrix can lead to a family of projection, pairs of projection matrices because now if you have a homography in the three-dimensional space like which means it is a 4 X 4 non-singular matrix homography of a 3D projective space.

$(P, P') \rightarrow F, (PH, P'H) \rightarrow F$

Then given a fundamental matrix F that exist a family of stereo setup or pairs of camera matrices, that we need to note. And this is one example of a family of matrices considering the reference camera is given always in the canonical form [I|0], then you can express this family given that fundamental matrix by its epipole and any arbitrary 3 vector and a arbitrary scalar constant, as you can see in that expression which has been shown here in this case. Then, given a camera matrices (P, P') and a corresponding pair of image points (x, x') it is possible to reconstruct the respective 3D scene point X.

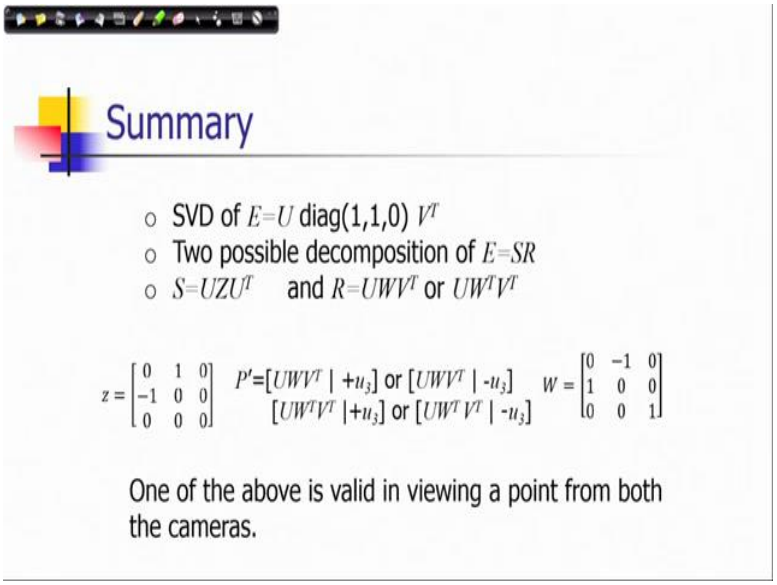
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We have discuss 3 approaches for this and one of them was using the triangulation geometry, the second one was using geometry preposition error is in non-linear optimization technique, and third one is algebraic form of representation of triangular triangulation approach. And that is how we can derive the structure.

Then, let us summarize also the facts regarding essential matrix which is a fundamental matrix of calibrated camera, and this is given in this structure. That is, if I considered a stereo system where projection matrices are given in this form that reference camera is canonical camera canonical matrix I 0 and the second camera is related by the rotation and translation of rotational axis and translation of R h. So, in that case essential matrix is given by those parameters only. And one of the properties of essential matrix is that two of its singular values are equal and the third one is 0.

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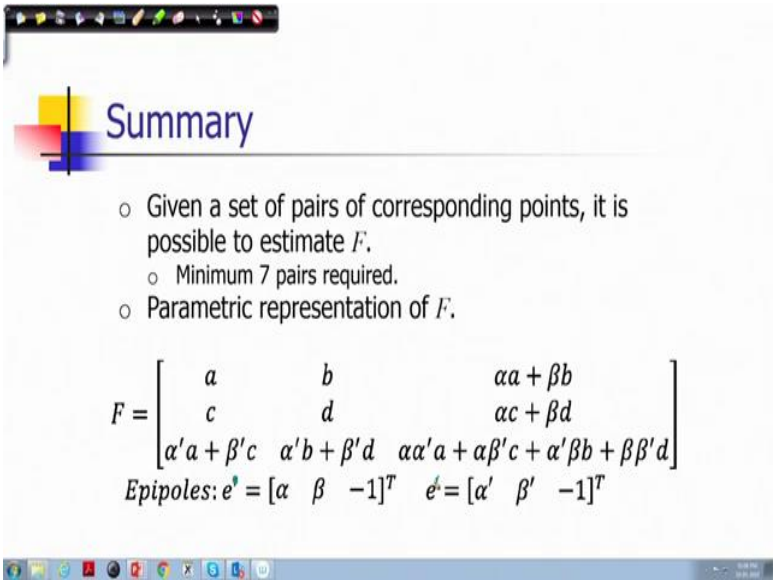
Then you can compute that it components t prime cross and R by matrix decomposition of E such that know E =SR, where S a skew symmetric matrix and R is orthogonal. And if you have this decomposition, then you can compute also its projection matrices which are shown here, that apply singular value decomposition of essential matrix. And, there could be two possible decomposition of this essential matrix which is shown here in this form that use the singular value decomposition matrix U and V and there are two special matrix Z and W which is shown here in this form.

Then the expression of S and R is given as you can see.

$$S = UZU^T, R = UWV^T \text{ or } UW^TV^T$$

And from there we can compute P'. So, note here this u3 is the third column of the matrix U. They are the singular value decomposition, which means this is a column vector corresponding to singular value 0. And one of the above is valid in viewing a point from both the cameras.

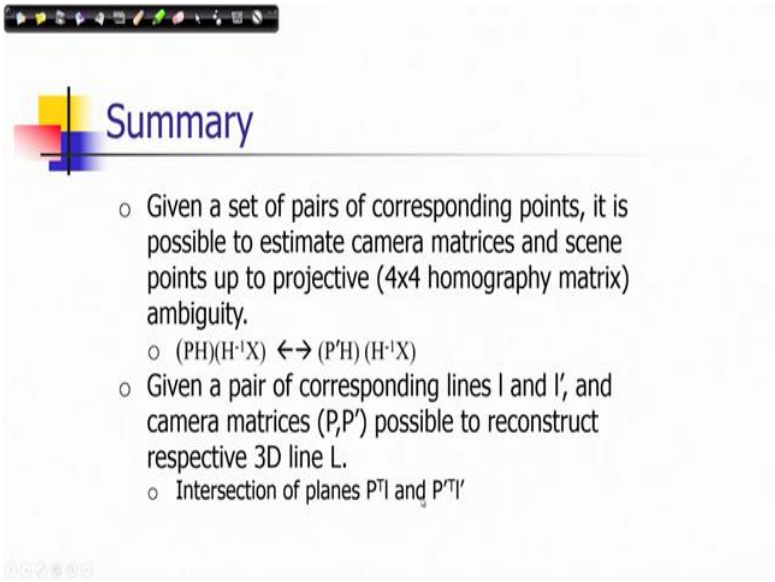
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And then we discussed also estimation of fundamental matrix. So, given a set of pairs of corresponding points it is possible to estimate fundamental matrix. Minimum 7 periods of points required, we have discussed one particular method where we have seen that how the eigenvectors corresponding to 0 eigenvalues they are used in this case, 0 eigenvalues, so eigenvectors of a system.

Then parametric representation of a fundamental matrix is given in this form. This is the particularly this from also contains the epipoles, you should note that know there is a little bit of ambiguity in representing e prime and e, e' should be e here and e should be e' here. This should be e'.

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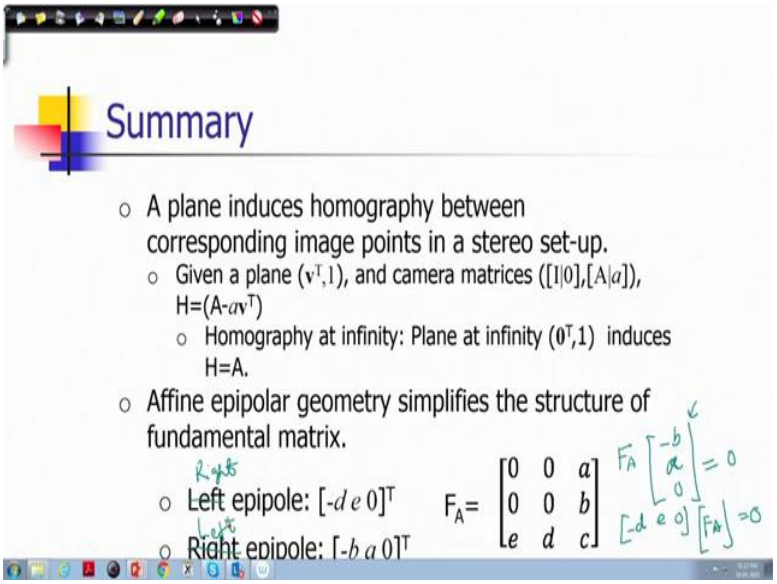


And given a set of pairs of corresponding points it is possible to estimate camera matrices and scene point up to projective ambiguity which is 4 X 4 homography matrix.

$$(PH)(H^{-1}X) \rightarrow (P'H)(H^{-1}X)$$

Then given a pair of corresponding lines l and l' and camera matrices P and P' it is possible to reconstruct respective 3D line L. We discuss that also. So, in this case you can compute the plain formed by the lines l and also camera centre we know that that is $P^T l$, and also the plane found by l' and camera centre of P' which is $P'^T l'$ and then two planes they intersect at a line which is that corresponding three-dimensional line, so that also you can reconstruct. So, this is what intersection of planes $P^T l$ and $P'^T l'$.

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Then, we discussed about a plane induced homography. So, a plane induces homography between corresponding image points in a stereo set up. So, this was the expression for the homography for a particular description of the camera matrices and the plane. So, plane described by say (v^T, l) transfers gives the directions of normal of the plane and camera matrices are given by [I|0] that is the canonical camera matrix of the reference camera and [A|a] that is a general camera matrix. Then, homography induced by that plane is given in into this form.

We also discussed about affine epipolar geometry which simplifies the structure of fundamental matrix.

$$F_A = \begin{matrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{matrix}$$

So, the epipoles they are given by is the right epipole which means the left epipole is at [-b a 0] and the right epipole is [-d e 0]. And there we are having the same confusion. Left epipole is the FA [-b a 0] that would be 0 which is the left epipole. So, it should be right epipole. And similarly [-d e 0]FA that should be equal to 0 which is actually may added. So, please note this correction once again here also.

Left epipole is the epipole at the left image plane and right epipole is the epipole at the right image plane. Thing with this we have come to the end of this particular topic and we will be discussing our next topic on feature detection in the next lecture.

Thank you very much.

Keywords: Affine epipolar geometry, fundamental matrix, left epipole, right epipole